

Cartesian Coordinates, Points, and Transformations

CIS - 600.455/655

Russell Taylor

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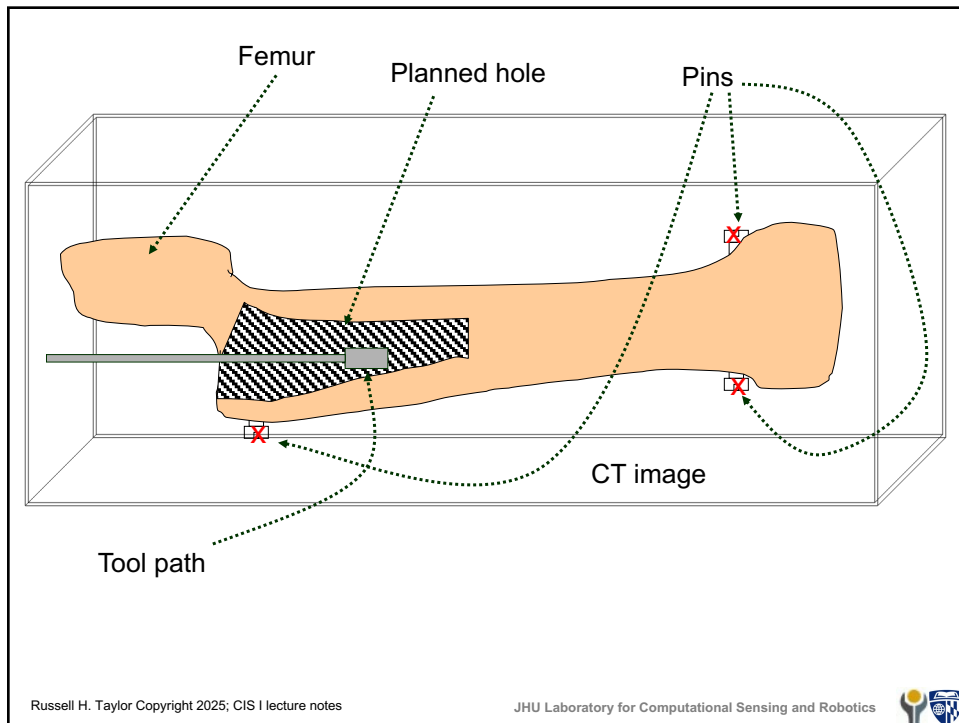


A diagram showing a robotic arm (orange) holding a human arm (light blue). The human arm is positioned horizontally. The robotic arm has a black gripper at the end, which is holding the human arm. The human arm is supported by a black frame. The gripper is a black rectangular block with a red 'X' on its side. The human arm is a light blue cylinder with a black band around the middle. The black frame consists of a vertical post and a horizontal bar. The background is white.

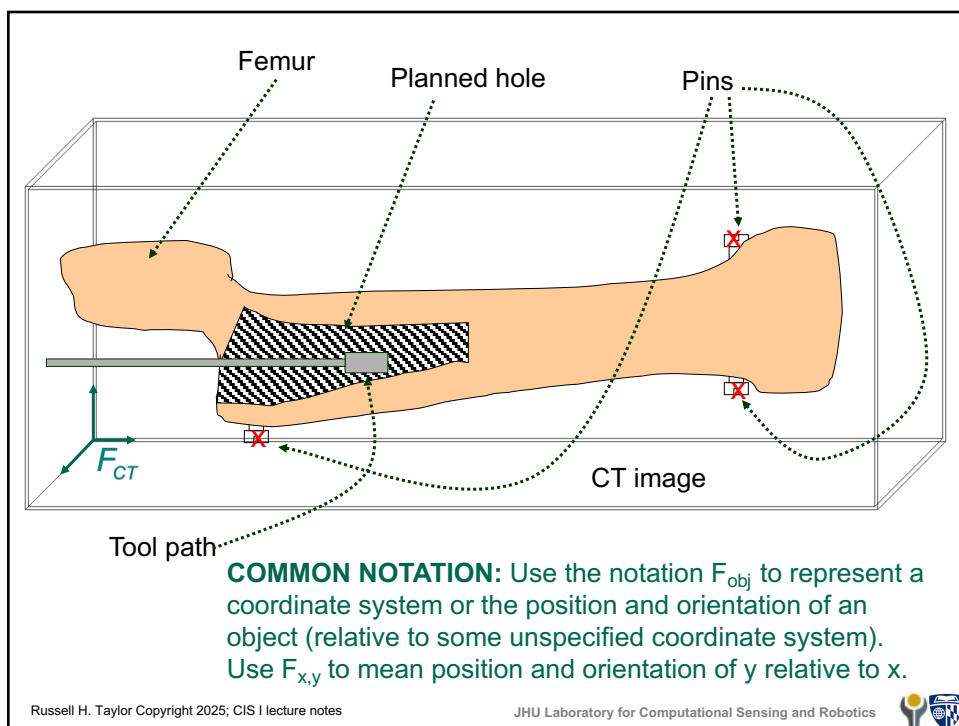
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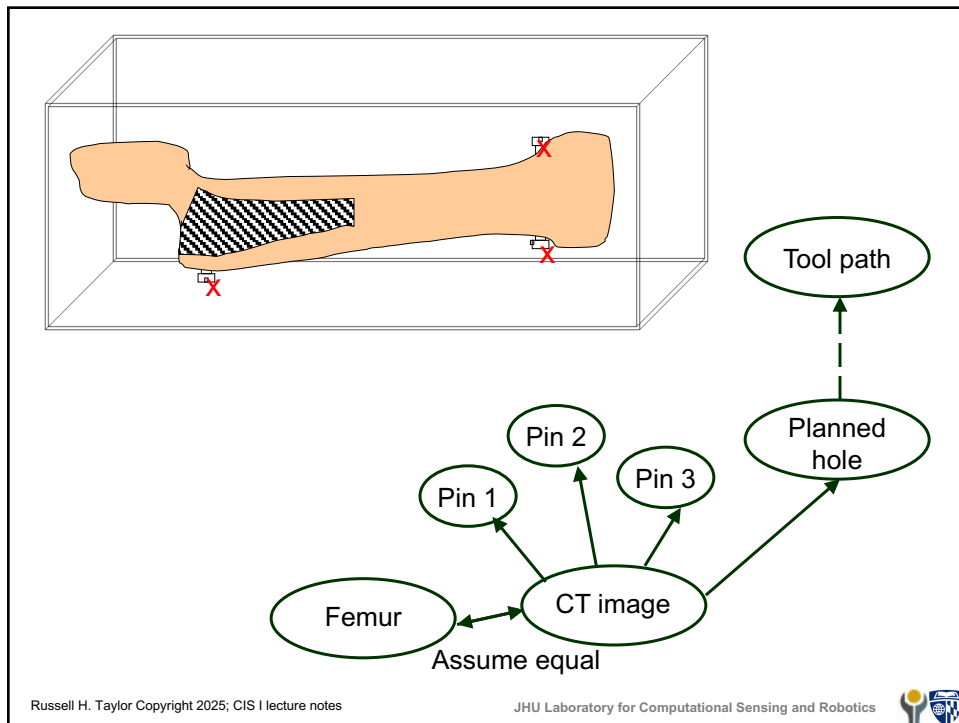
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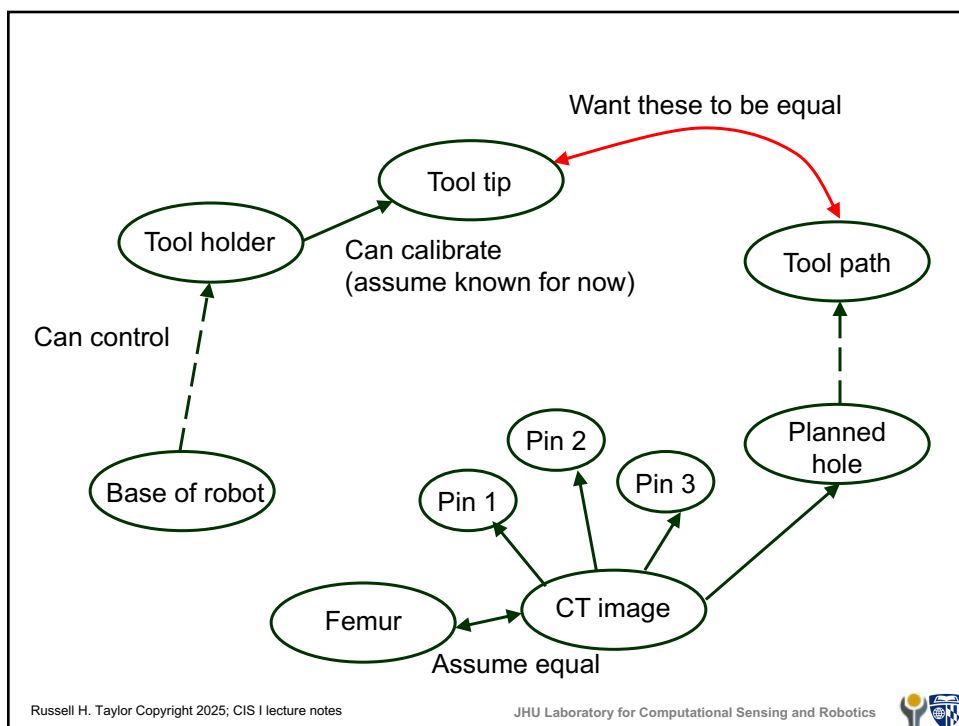
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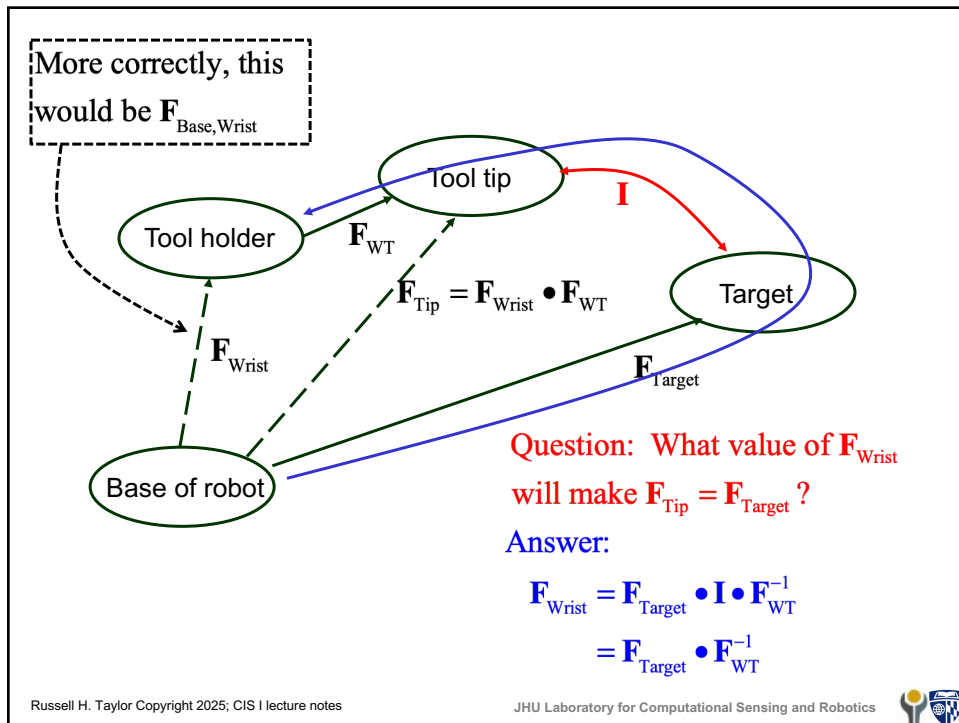
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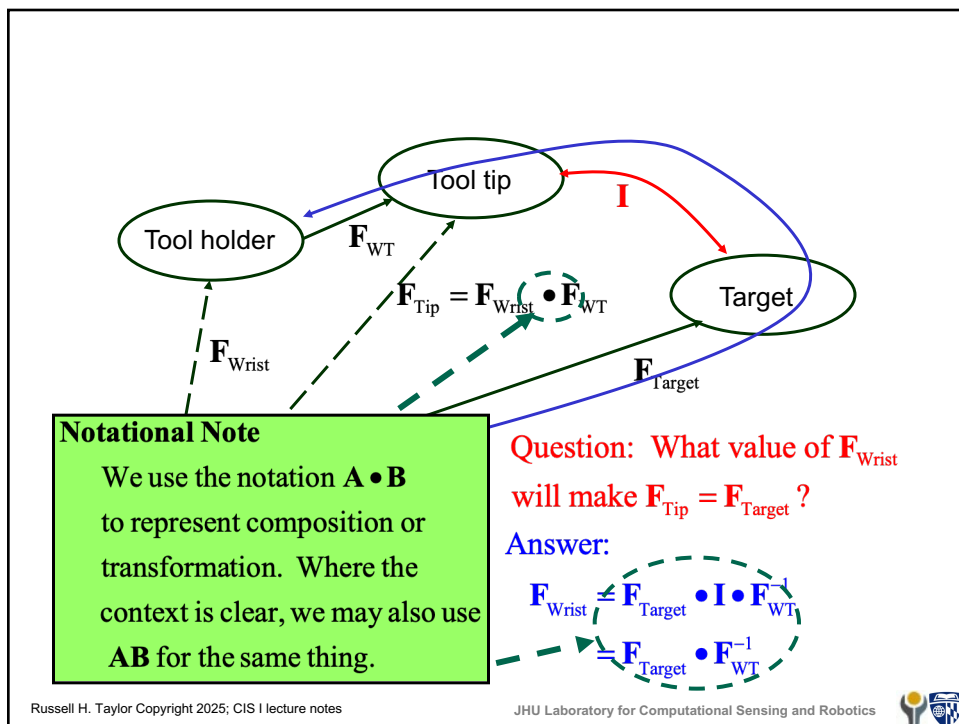
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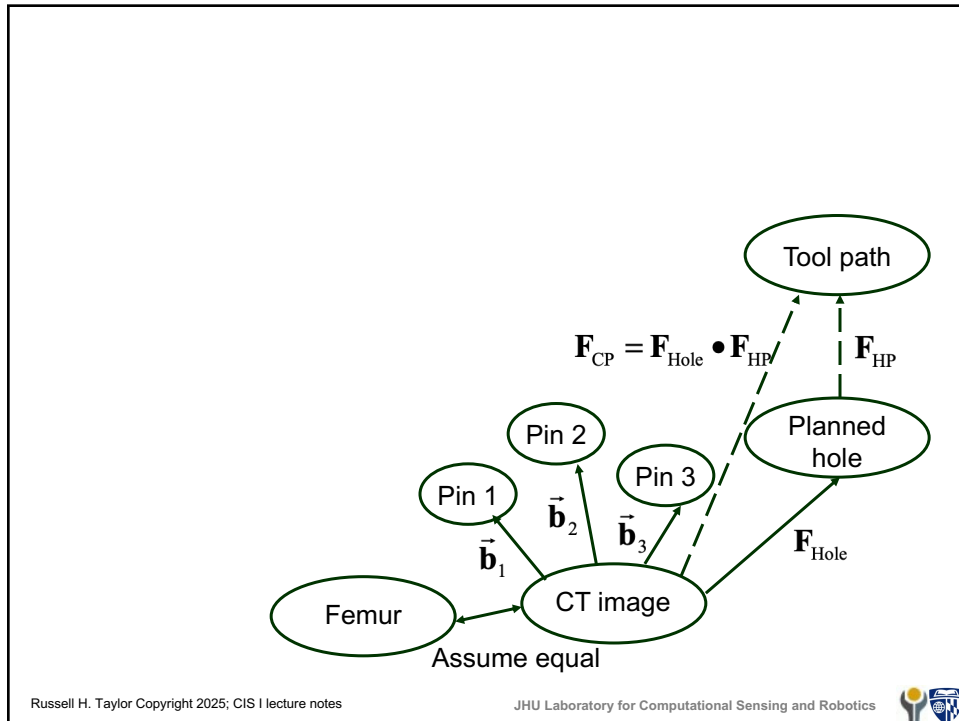
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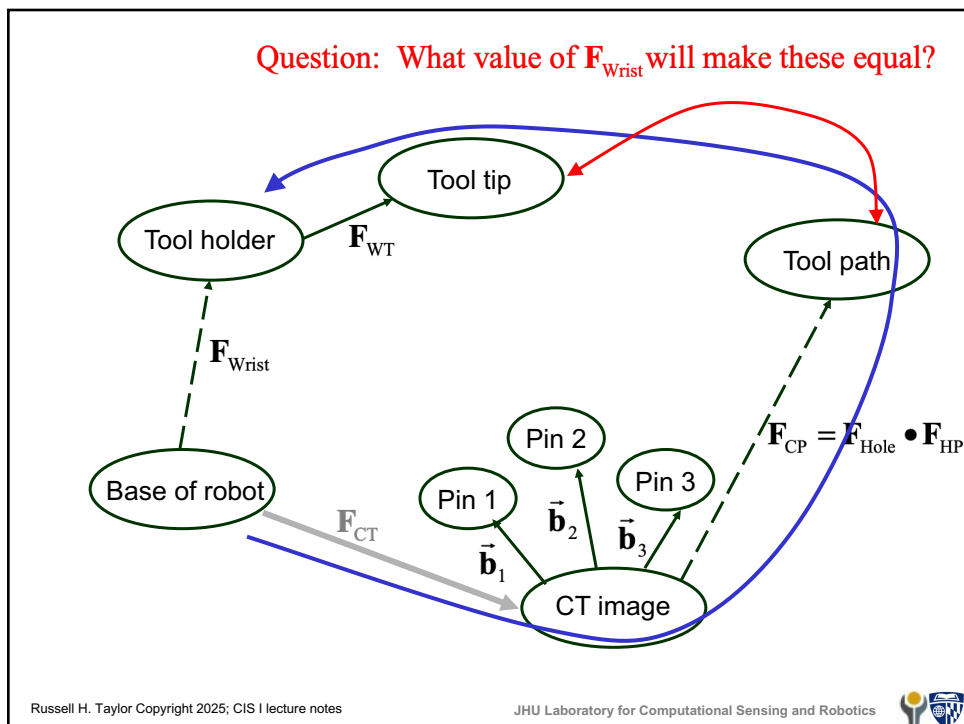
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9

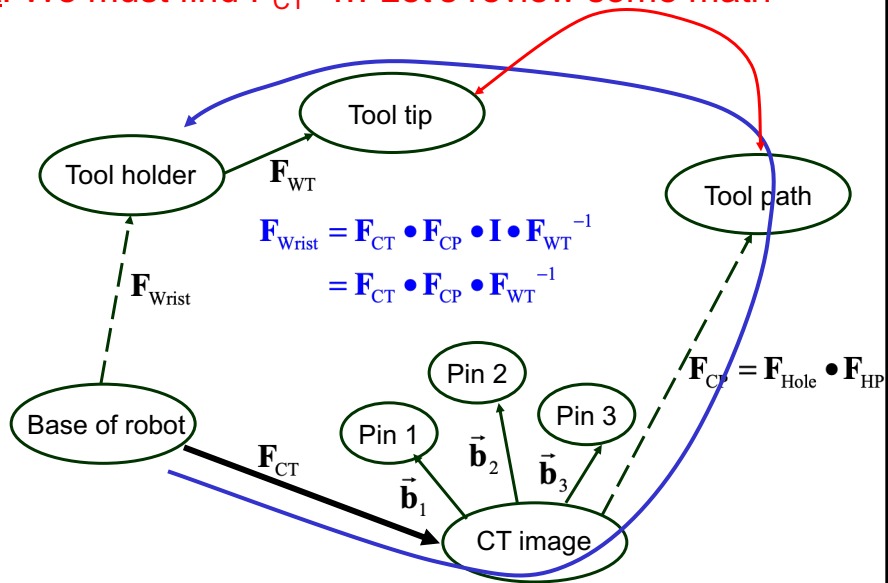


10



11

But: We must find F_{CT} ... Let's review some math



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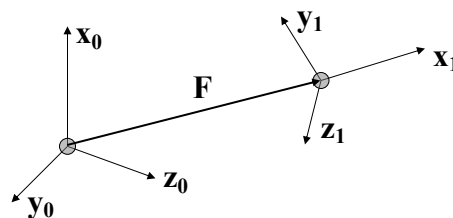
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12

Coordinate Frame Transformation

$$F = [R, p]$$

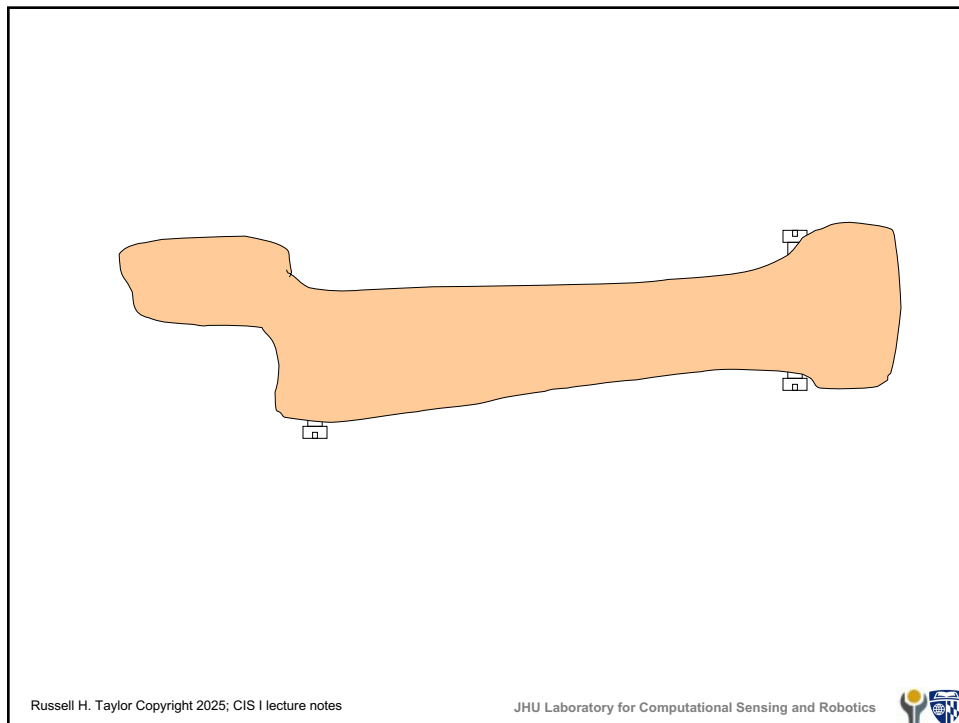


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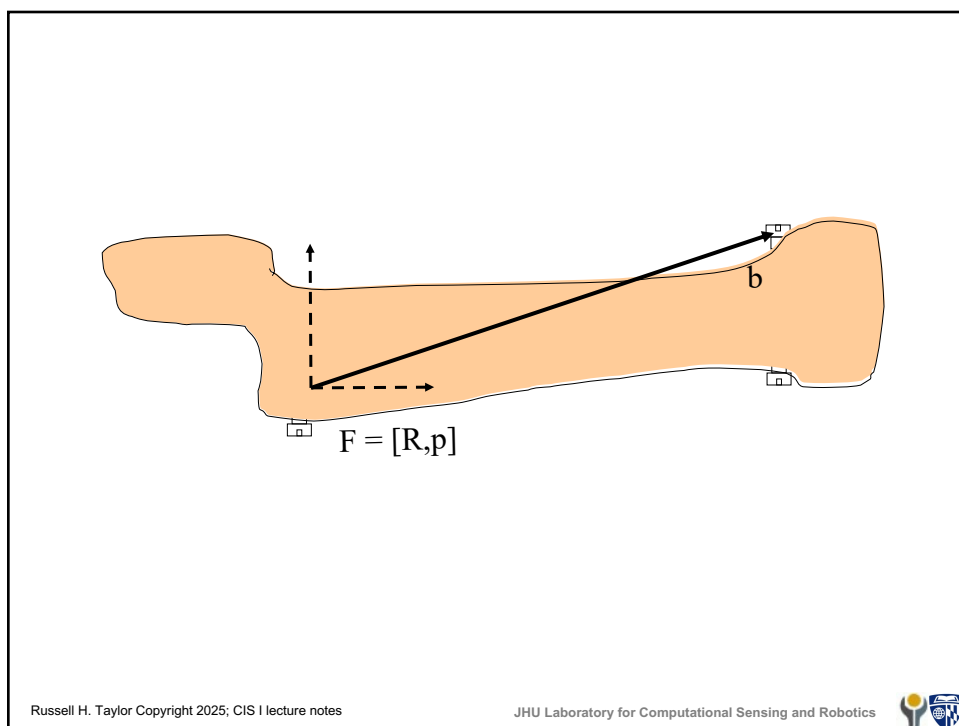
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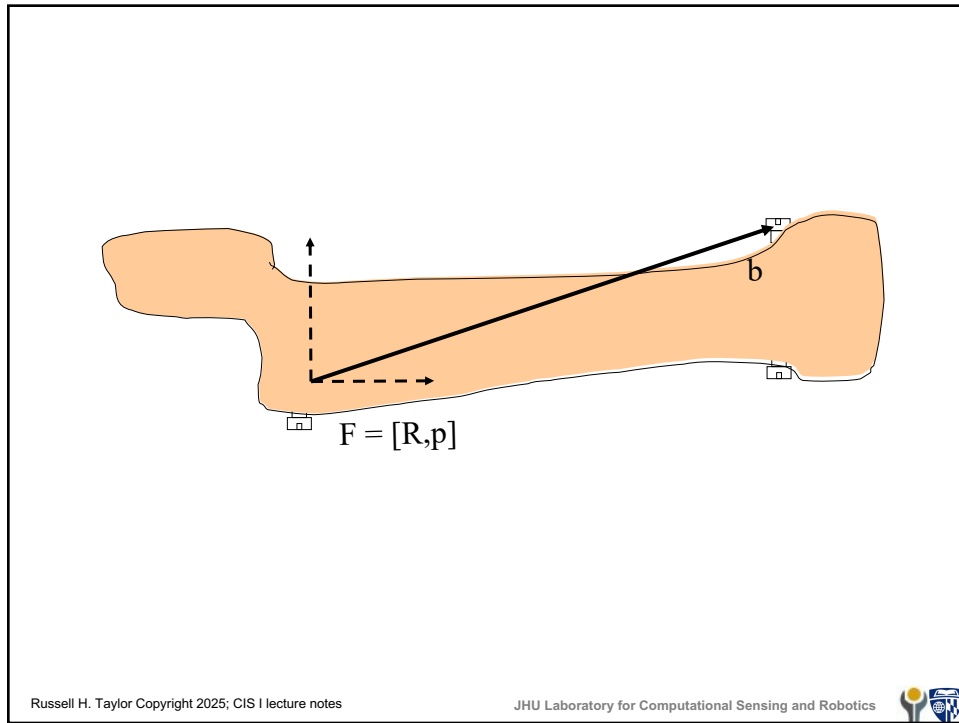
13



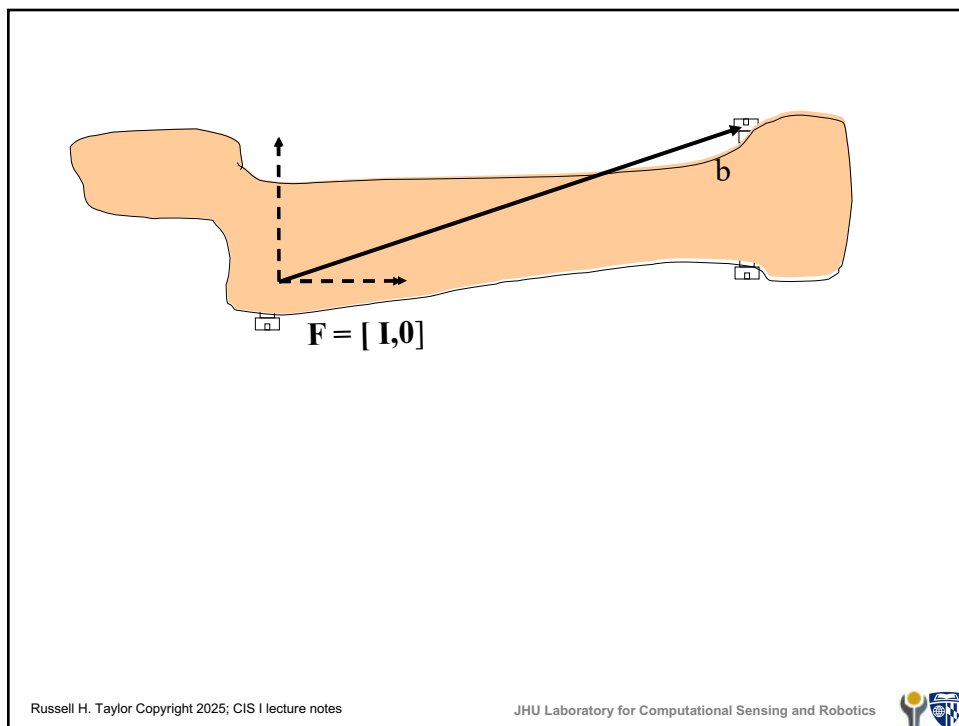
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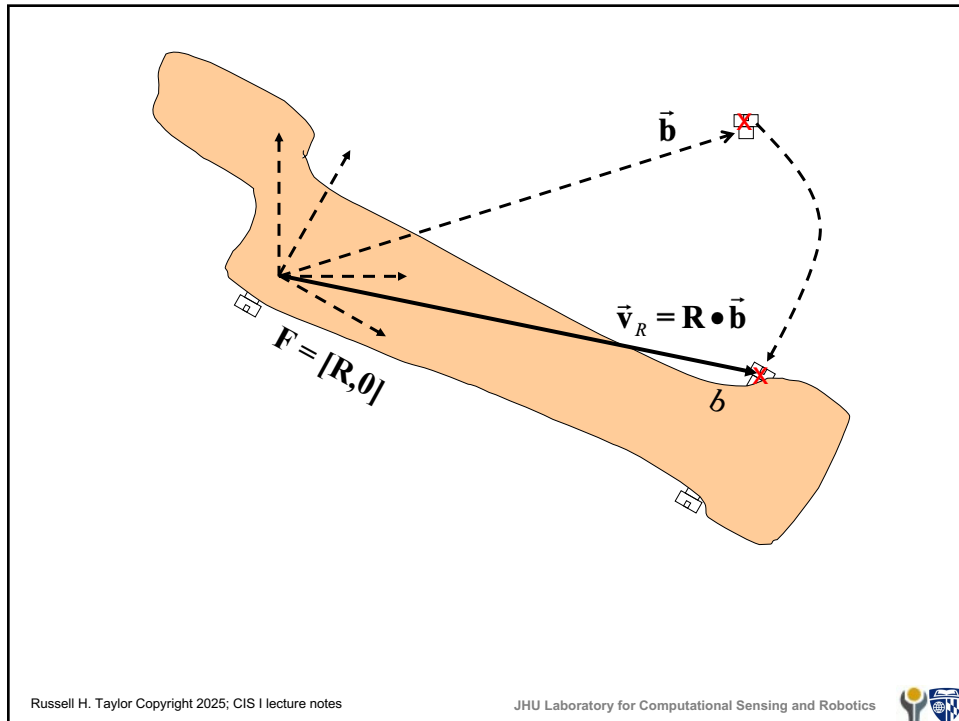
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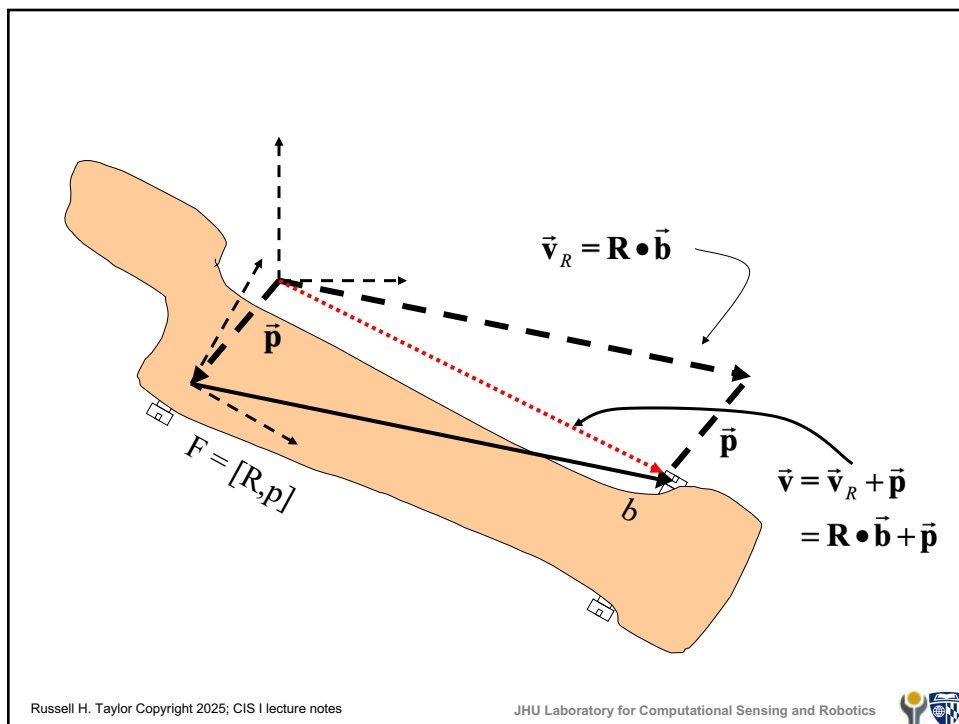
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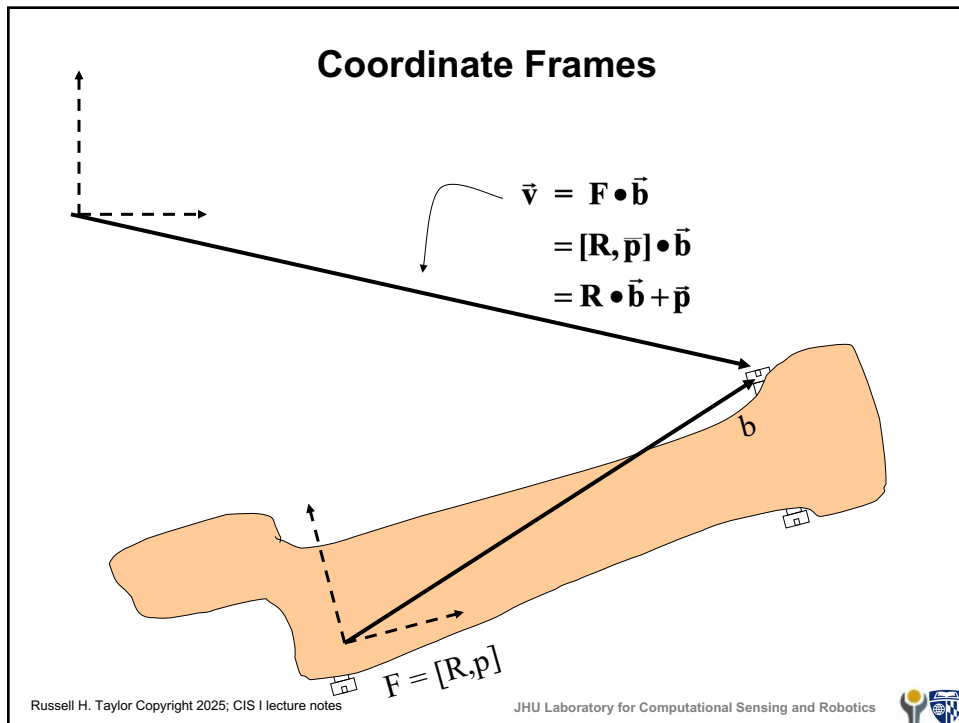
17



18



19



20

Forward and Inverse Frame Transformations

Forward	Inverse
$\mathbf{F} = [\mathbf{R}, \vec{p}]$	$\mathbf{F}^{-1} \vec{v} = \vec{b}$
	$\vec{b} = \mathbf{R}^{-1} \bullet (\vec{v} - \vec{p})$
	$= \mathbf{R}^{-1} \bullet \vec{v} - \mathbf{R}^{-1} \bullet \vec{p}$
$\vec{v} = \mathbf{F} \bullet \vec{b}$	
$= [\mathbf{R}, \vec{p}] \bullet \vec{b}$	
$= \mathbf{R} \bullet \vec{b} + \vec{p}$	
	$\mathbf{F}^{-1} = [\mathbf{R}^{-1}, -\mathbf{R}^{-1} \bullet \vec{p}]$

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21

Composition

Assume $\mathbf{F}_1 = [\mathbf{R}_1, \vec{\mathbf{p}}_1]$, $\mathbf{F}_2 = [\mathbf{R}_2, \vec{\mathbf{p}}_2]$

Then

$$\begin{aligned}\mathbf{F}_1 \bullet \mathbf{F}_2 \bullet \vec{\mathbf{b}} &= \mathbf{F}_1 \bullet (\mathbf{F}_2 \bullet \vec{\mathbf{b}}) \\ &= \mathbf{F}_1 \bullet (\mathbf{R}_2 \bullet \vec{\mathbf{b}} + \vec{\mathbf{p}}_2) \\ &= [\mathbf{R}_1, \vec{\mathbf{p}}_1] \bullet (\mathbf{R}_2 \bullet \vec{\mathbf{b}} + \vec{\mathbf{p}}_2) \\ &= \mathbf{R}_1 \bullet (\mathbf{R}_2 \bullet \vec{\mathbf{b}} + \vec{\mathbf{p}}_2) + \vec{\mathbf{p}}_1 \\ &= \mathbf{R}_1 \bullet \mathbf{R}_2 \bullet \vec{\mathbf{b}} + \mathbf{R}_1 \bullet \vec{\mathbf{p}}_2 + \vec{\mathbf{p}}_1 \\ &= [\mathbf{R}_1 \bullet \mathbf{R}_2, \mathbf{R}_1 \bullet \vec{\mathbf{p}}_2 + \vec{\mathbf{p}}_1] \bullet \vec{\mathbf{b}}\end{aligned}$$

So

$$\begin{aligned}\mathbf{F}_1 \bullet \mathbf{F}_2 &= [\mathbf{R}_1, \vec{\mathbf{p}}_1] \bullet [\mathbf{R}_2, \vec{\mathbf{p}}_2] \\ &= [\mathbf{R}_1 \bullet \mathbf{R}_2, \mathbf{R}_1 \bullet \vec{\mathbf{p}}_2 + \vec{\mathbf{p}}_1]\end{aligned}$$

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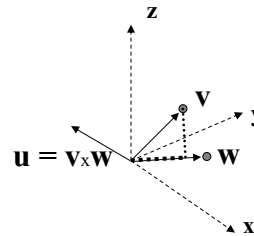


22

Vectors

$$\vec{\mathbf{v}}_{col} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \quad \leftarrow \text{We usually will assume this}$$

$$\vec{\mathbf{v}}_{row} = \begin{bmatrix} v_x & v_y & v_z \end{bmatrix}$$



$$\text{length: } \|\vec{\mathbf{v}}\| = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\text{dot product: } a = \vec{\mathbf{v}} \cdot \vec{\mathbf{w}} = (v_x w_x + v_y w_y + v_z w_z) = \|\vec{\mathbf{v}}\| \|\vec{\mathbf{w}}\| \cos \theta$$

$$\text{cross product: } \vec{\mathbf{u}} = \vec{\mathbf{v}} \times \vec{\mathbf{w}} = \begin{bmatrix} v_y w_z - v_z w_y \\ v_z w_x - v_x w_z \\ v_x w_y - v_y w_x \end{bmatrix}, \|\vec{\mathbf{u}}\| = \|\vec{\mathbf{v}}\| \|\vec{\mathbf{w}}\| \sin \theta$$

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Slide acknowledgment: Sarah Graham and Andy Bzostek

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23

Matrix representation of cross product operator

Define

$$\hat{\vec{a}} \triangleq skew(\vec{a}) \triangleq \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}$$

Then

$$\vec{a} \times \vec{v} = skew(\vec{a}) \bullet \vec{v}$$



25

Points and Lines

$$\vec{e} = (\vec{q} - \vec{c}) - (\vec{n} \cdot (\vec{q} - \vec{c})\vec{n})$$

$$\vec{p} = \vec{c} + \lambda \vec{d}$$

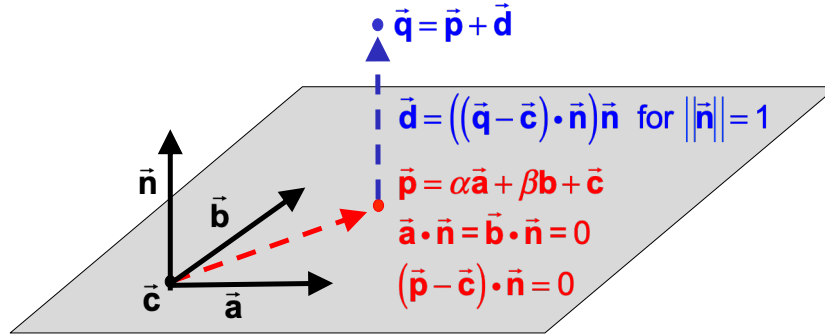
$$\lambda = \frac{\|\vec{p} - \vec{c}\|}{\|\vec{d}\|} = \|\vec{p} - \vec{c}\| \text{ for } \|\vec{d}\| = 1$$

$$\vec{a} \cdot (\vec{p} - \vec{c}) = \vec{b} \cdot (\vec{p} - \vec{c}) = 0$$



26

Points and Planes



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27

Rotations: Some Notation

$Rot(\vec{a}, \alpha) =$ Rotation by angle α about axis $\vec{a}$

$\mathbf{R}_{\vec{a}}(\alpha) =$ Rotation by angle α about axis $\vec{a}$

$\mathbf{R}(\vec{a}) = Rot(\vec{a}, \|\vec{a}\|)$

$\mathbf{R}_{xyz}(\alpha, \beta, \gamma) = \mathbf{R}(\vec{x}, \alpha) \cdot \mathbf{R}(\vec{y}, \beta) \cdot \mathbf{R}(\vec{z}, \gamma)$

$\mathbf{R}_{zyz}(\alpha, \beta, \gamma) = \mathbf{R}(\vec{z}, \alpha) \cdot \mathbf{R}(\vec{y}, \beta) \cdot \mathbf{R}(\vec{z}, \gamma)$

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29

Rotations: A few useful facts

$$Rot(\vec{s}\vec{a}, \alpha) \bullet \vec{a} = \vec{a} \quad \text{and} \quad \|Rot(\vec{a}, \alpha) \bullet \vec{b}\| = \|\vec{b}\|$$

$$Rot(\vec{a}, \alpha) = Rot(\hat{a}, \alpha) \quad \text{where} \quad \hat{a} = \frac{\vec{a}}{\|\vec{a}\|}$$

NOTE: Unless otherwise stated, we usually will assume that $\vec{a}$ in $Rot(\vec{a}, \theta)$ is a unit vector. I.e., $\|\vec{a}\| = 1$.

$$Rot(\vec{a}, \alpha) \bullet Rot(\vec{a}, \beta) = Rot(\vec{a}, \alpha + \beta)$$

$$Rot(\vec{a}, \alpha)^{-1} = Rot(\vec{a}, -\alpha)$$

$$Rot(\vec{a}, 0) \bullet \vec{b} = \vec{b} \quad \text{i.e.,} \quad Rot(\vec{a}, 0) = \mathbf{I}_{Rot} = \text{the identity rotation}$$

$$Rot(\hat{a}, \alpha) \bullet \vec{b} = (\hat{a} \bullet \vec{b})\hat{a} + Rot(\hat{a}, \alpha) \bullet (\vec{b} - (\hat{a} \bullet \vec{b})\hat{a})$$

$$Rot(\hat{a}, \alpha) \bullet Rot(\hat{b}, \beta) = Rot(\hat{b}, \beta) \bullet Rot(Rot(\hat{b}, -\beta) \bullet \hat{a}, \alpha)$$

$$Rot(\hat{a}, \alpha) \bullet \mathbf{R}_\beta = \mathbf{R}_\beta \bullet Rot(\mathbf{R}_\beta^{-1} \bullet \hat{a}, \alpha)$$

$$\mathbf{R}_\alpha \bullet Rot(\hat{b}, \beta) = Rot(\mathbf{R}_\alpha \bullet \hat{b}, \beta) \bullet \mathbf{R}_\alpha$$

Especially useful

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31

Rotations: more facts

If $\vec{v} = [v_x, v_y, v_z]^T$ then a rotation $\mathbf{R} \bullet \vec{v}$ may be described in terms of the effects of $\mathbf{R}$ on orthogonal unit vectors, $\vec{e}_x = [1, 0, 0]^T$, $\vec{e}_y = [0, 1, 0]^T$, $\vec{e}_z = [0, 0, 1]^T$

$$\mathbf{R} \bullet \vec{v} = v_x \vec{r}_x + v_y \vec{r}_y + v_z \vec{r}_z$$

where

$$\vec{r}_x = \mathbf{R} \bullet \vec{e}_x$$

$$\vec{r}_y = \mathbf{R} \bullet \vec{e}_y$$

$$\vec{r}_z = \mathbf{R} \bullet \vec{e}_z$$

Note that rotation doesn't affect inner products

$$(\mathbf{R} \bullet \vec{b}) \bullet (\mathbf{R} \bullet \vec{c}) = \vec{b} \bullet \vec{c}$$

or lengths of vectors

$$\|\mathbf{R} \bullet \vec{v}\| = \|\vec{v}\|$$

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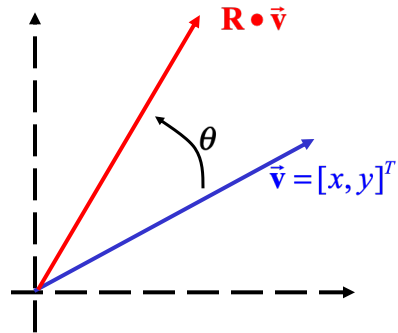
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32

Rotations in the plane

$$\begin{aligned}\mathbf{R} \cdot \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}\end{aligned}$$



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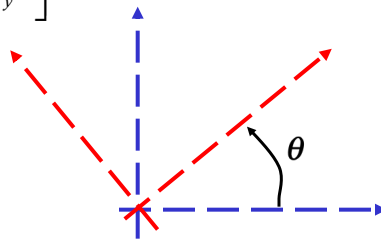
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33

Rotations in the plane

$$\begin{aligned}\mathbf{R} \cdot \begin{bmatrix} \vec{e}_x & \vec{e}_y \end{bmatrix} &= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R} \cdot \vec{e}_x & \mathbf{R} \cdot \vec{e}_y \end{bmatrix} \\ &= \begin{bmatrix} \vec{r}_x & \vec{r}_y \end{bmatrix}\end{aligned}$$



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34

3D Rotation Matrices

$$\begin{aligned}
 \mathbf{R} \bullet [\vec{\mathbf{e}}_x \quad \vec{\mathbf{e}}_y \quad \vec{\mathbf{e}}_z] &= [\mathbf{R} \bullet \vec{\mathbf{e}}_x \quad \mathbf{R} \bullet \vec{\mathbf{e}}_y \quad \mathbf{R} \bullet \vec{\mathbf{e}}_z] \\
 &= [\vec{\mathbf{r}}_x \quad \vec{\mathbf{r}}_y \quad \vec{\mathbf{r}}_z] \\
 \mathbf{R}^T \bullet \mathbf{R} &= \begin{bmatrix} \hat{\mathbf{r}}_x^T \\ \hat{\mathbf{r}}_y^T \\ \hat{\mathbf{r}}_z^T \end{bmatrix} \bullet \begin{bmatrix} \vec{\mathbf{r}}_x & \vec{\mathbf{r}}_y & \vec{\mathbf{r}}_z \end{bmatrix} \\
 &= \begin{bmatrix} \vec{\mathbf{r}}_x^T \bullet \vec{\mathbf{r}}_x & \vec{\mathbf{r}}_x^T \bullet \vec{\mathbf{r}}_y & \vec{\mathbf{r}}_x^T \bullet \vec{\mathbf{r}}_z \\ \vec{\mathbf{r}}_y^T \bullet \vec{\mathbf{r}}_x & \vec{\mathbf{r}}_y^T \bullet \vec{\mathbf{r}}_y & \vec{\mathbf{r}}_y^T \bullet \vec{\mathbf{r}}_z \\ \vec{\mathbf{r}}_z^T \bullet \vec{\mathbf{r}}_x & \vec{\mathbf{r}}_z^T \bullet \vec{\mathbf{r}}_y & \vec{\mathbf{r}}_z^T \bullet \vec{\mathbf{r}}_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

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35

Properties of Rotation Matrices

Inverse of a Rotation Matrix equals its transpose:

$$\mathbf{R}^{-1} = \mathbf{R}^T$$

$$\mathbf{R}^T \mathbf{R} = \mathbf{R} \mathbf{R}^T = \mathbf{I}$$

The Determinant of a Rotation matrix is equal to +1:

$$\det(\mathbf{R}) = +1$$

Any Rotation can be described by consecutive rotations about the three primary axes, x, y, and z, e.g.,

$$\mathbf{R} = \mathbf{R}_{x,\theta} \mathbf{R}_{y,\phi} \mathbf{R}_{z,\psi}$$

$$\mathbf{R} = \mathbf{R}_{z,\theta} \mathbf{R}_{y,\phi} \mathbf{R}_{x,\psi}$$

$$\mathbf{R} = \mathbf{R}_{z,\theta} \mathbf{R}_{x,\phi} \mathbf{R}_{y,\psi}$$

We usually will use this one

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36

Canonical 3D Rotation Matrices

Note: Right-Handed Coordinate System

$$\mathbf{R}_{\vec{x}}(\theta) = \text{Rot}(\vec{x}, \theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$\mathbf{R}_{\vec{y}}(\theta) = \text{Rot}(\vec{y}, \theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

$$\mathbf{R}_{\vec{z}}(\theta) = \text{Rot}(\vec{z}, \theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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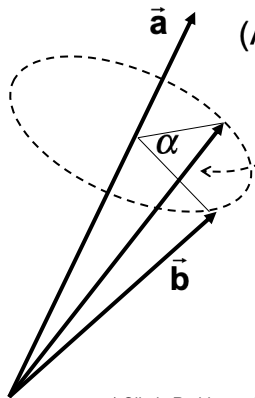


37

Axis-angle Representations of Rotations

Rodrigues' Formula (1840)*

Rotation of a vector $\vec{b}$ by angle α about axis $\vec{a}$
(Assumes that $\vec{a}$ is a unit vector, $\|\vec{a}\| = 1$)



$$\begin{aligned} \vec{c} &= \text{Rot}(\vec{a}, \alpha) \cdot \vec{b} \\ &= \vec{b} \cos \alpha + \vec{a} \times \vec{b} \sin \alpha + \vec{a}(\vec{a} \cdot \vec{b})(1 - \cos \alpha) \end{aligned}$$

In matrix form this is $\vec{c} = \mathbf{R} \cdot \vec{b}$ where

$$\mathbf{R} = (\cos \alpha) \mathbf{I} + (\sin \alpha) \text{skew}(\vec{a}) + (1 - \cos \alpha) \vec{a} \cdot \vec{a}^T$$

* Olinde Rodrigues, "Des lois géométriques qui régissent les déplacements d'un système solide dans l'espace, et de la variation des coordonnées provenant de ces déplacements considérés indépendants des causes qui peuvent les produire", *Journal de Mathématiques Pures et Appliquées* 5 (1840), 380–440. (http://sites.mathdoc.fr/JMPA/PDF/JMPA_1840_1_5_A39_0.pdf)

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38

Exponential representation

Consider the matrix exponential of $\text{skew}(\theta \vec{n})$, where $\|\vec{n}\| = 1$. We have

$$e^{\text{skew}(\vec{n})\theta} = \mathbf{I} + \theta \text{skew}(\vec{n}) + \frac{\theta^2}{2!} \text{skew}(\vec{n})^2 + \dots + \frac{\theta^k}{k!} \text{skew}(\vec{n})^k + \dots$$

Since $\text{skew}(\vec{n})^3 = -\text{skew}(\vec{n})$, by doing some manipulation, you can show

$$\begin{aligned} e^{\text{skew}(\vec{n})\theta} &= \mathbf{I} + \text{skew}(\vec{n})\sin\theta + \text{skew}(\vec{n})^2(1 - \cos\theta) \\ &= \mathbf{I} + \text{skew}(\vec{n})\sin\theta + (\vec{n} \bullet \vec{n}^T - \mathbf{I})(1 - \cos\theta) \\ &= \mathbf{I}\cos\theta + \text{skew}(\vec{n})\sin\theta + \vec{n} \bullet \vec{n}^T(1 - \cos\theta) \end{aligned}$$

which is just Rodrigues' formula for $\text{Rot}(\vec{n}, \theta)$.

Note that for small θ , this reduces to

$$\text{Rot}(\vec{n}, \theta) \approx \mathbf{I} + \text{skew}(\theta \vec{n})$$



Cayley Transform Representation

Consider the rotation $\text{Rot}(\vec{n}, \theta)$ and define

$$\vec{a} = \left(\tan \frac{\theta}{2} \right) \vec{n}$$

$$\mathbf{A} = \text{skew}(\vec{a})$$

Then,

$$\mathbf{R} = (\mathbf{I} - \mathbf{A})^{-1}(\mathbf{I} + \mathbf{A})$$

gives the rotation matrix corresponding to $\text{Rot}(\vec{n}, \theta)$. Similarly, given $\mathbf{R}$,

$$\mathbf{A} = (\mathbf{R} - \mathbf{I})(\mathbf{I} + \mathbf{R})^{-1}$$

gives the elements of $\text{skew}(\vec{a})$ and hence for $\vec{a}$.

Note: The above relations require that $\theta \neq \pm\pi$.



Quaternion Representation

Invented by Hamilton in 1843. Can be thought of as

$$\begin{aligned} \text{4 elements:} \quad & \mathbf{q} = [q_0, q_1, q_2, q_3] \\ \text{scalar \& vector:} \quad & \mathbf{q} = s + \vec{\mathbf{v}} = [s, \vec{\mathbf{v}}] \\ \text{Complex number:} \quad & \mathbf{q} = q_0 + q_1 i + q_2 j + q_3 k \\ & \text{where } i^2 = j^2 = k^2 = i j k = -1 \end{aligned}$$

Properties:

$$\begin{aligned} \text{Linearity:} \quad & \lambda \mathbf{q}_1 + \mu \mathbf{q}_2 = [\lambda s_1 + \mu s_2, \lambda \vec{\mathbf{v}}_1 + \mu \vec{\mathbf{v}}_2] \\ \text{Conjugate:} \quad & \mathbf{q}^* = s - \vec{\mathbf{v}} = [s, -\vec{\mathbf{v}}] \\ \text{Product:} \quad & \mathbf{q}_1 \circ \mathbf{q}_2 = [s_1 s_2 - \vec{\mathbf{v}}_1 \cdot \vec{\mathbf{v}}_2, s_1 \vec{\mathbf{v}}_2 + s_2 \vec{\mathbf{v}}_1 + \vec{\mathbf{v}}_1 \times \vec{\mathbf{v}}_2] \\ \text{Transform vector:} \quad & \mathbf{q} \circ \vec{\mathbf{p}} = \mathbf{q} \circ [0, \vec{\mathbf{p}}] \circ \mathbf{q}^* \\ \text{Norm:} \quad & \|\mathbf{q}\| = \sqrt{s^2 + \vec{\mathbf{v}} \cdot \vec{\mathbf{v}}} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} \end{aligned}$$

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42

Unit Quaternions

We can associate a rotation by angle θ about an axis $\vec{\mathbf{n}}$ with the unit quaternion:

$$\text{Rot}(\vec{\mathbf{n}}, \theta) \Leftrightarrow [\cos(\theta/2), \sin(\theta/2)\vec{\mathbf{n}}]$$

Quaternion to XYZ Euler angles: if $(s, \vec{\mathbf{v}}) \Leftrightarrow \text{Rot}_{xyz}(\alpha, \beta, \gamma)$ then

$$\alpha = \vec{\mathbf{v}} \cdot \vec{\mathbf{e}}_x \quad \beta = \vec{\mathbf{v}} \cdot \vec{\mathbf{e}}_y \quad \gamma = \vec{\mathbf{v}} \cdot \vec{\mathbf{e}}_z \quad \text{so } \vec{\mathbf{v}} = [\alpha, \beta, \gamma]^T$$

Exercise: Demonstrate these relationships.

Hint: substitute and show that

$$\text{Rot}((\vec{\mathbf{n}}, \theta) \cdot \vec{\mathbf{p}} = [\cos(\theta/2), \sin(\theta/2)\vec{\mathbf{n}}] \circ [0, \vec{\mathbf{p}}] \circ [\cos(\theta/2), -\sin(\theta/2)\vec{\mathbf{n}}]$$

yields Rodriguez formula

Hint: And also rely on the relationships

$$s = \cos(\theta/2); \quad \mathbf{v}_x = s \times \cos(\alpha/2); \quad \mathbf{v}_y = s \times \cos(\beta/2); \quad \mathbf{v}_z = s \times \cos(\gamma/2)$$

Ref: https://en.wikipedia.org/wiki/Conversion_between_quaternions_and_Euler_angles

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43

A bit more on quaternions

Exercise: show by substitution that the various formulations for quaternions are equivalent

A few web references:

<http://mathworld.wolfram.com/Quaternion.html>

<http://en.wikipedia.org/wiki/Quaternion>

http://en.wikipedia.org/wiki/Quaternions_and_spatial_rotation

<http://www.euclideanspace.com/maths/algebra/realNormedAlgebra/quaternions/index.htm>

CAUTION: Different software packages are not always consistent in the order of elements if a quaternion is represented as a 4 element vector. Some put the scalar part first, others (including cist libraries) put it last.



Note on the difference between two rotations

One often wants to consider the "difference" between two rotations

$$\mathbf{R}_{12} = \mathbf{R}_1^{-1} \cdot \mathbf{R}_2$$

In these cases, it is useful to consider an axis-angle representation

$$\mathbf{R}_{12} = \text{Rot}(\bar{\mathbf{n}}_{12}, \theta_{12})$$

Given $\mathbf{R}_{12}$, there are several ways to extract $\bar{\mathbf{n}}_{12}$ and $\bar{\theta}_{12}$. For example,

you can use the Cayley formula to compute $\bar{\mathbf{a}}_{12} = \tan(\theta_{12} / 2) \bar{\mathbf{n}}_{12}$ from

$$sk(\bar{\mathbf{a}}_{12}) = (\mathbf{R}_{12} - \mathbf{I})(\mathbf{I} + \mathbf{R}_{12})^{-1}$$

and then

$$\theta_{12} = 2 \arctan(\|\bar{\mathbf{a}}_{12}\|) \quad \text{and} \quad \bar{\mathbf{n}}_{12} = \bar{\mathbf{a}}_{12} / \|\bar{\mathbf{a}}_{12}\|$$

Also, if the rotations are very close to each other so that $\bar{\theta}_{12}$ is small, then

$$\mathbf{R}_{12} \approx \mathbf{I} + sk(\bar{\alpha}_{12})$$

So

$$\theta_{12} \approx \|\bar{\alpha}_{12}\| \quad \text{and} \quad \bar{\mathbf{n}}_{12} \approx \bar{\alpha}_{12} / \theta_{12}$$

This last relationship is very useful in reporting things like registration error since the elements $\bar{\alpha}_{12}$ are small rotations about the x,y,z axes



Homogeneous Coordinates

- Widely used in graphics, geometric calculations
- Represent 3D vector as 4D quantity
- For our current purposes, we will keep the “scale” $s = 1$

$$\vec{\mathbf{V}} \equiv \begin{bmatrix} xS \\ yS \\ zS \\ S \end{bmatrix} \cong \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

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46

Representing Frame Transformations as Matrices

$$\mathbf{v} + \mathbf{p} \rightarrow \begin{bmatrix} 1 & 0 & 0 & \mathbf{p}_x \\ 0 & 1 & 0 & \mathbf{p}_y \\ 0 & 0 & 1 & \mathbf{p}_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{v}_x \\ \mathbf{v}_y \\ \mathbf{v}_z \\ 1 \end{bmatrix} = [\mathbf{I}, \tilde{\mathbf{p}}] \bullet \mathbf{v}$$

$$\mathbf{R} \bullet \mathbf{v} \rightarrow \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ 1 \end{bmatrix}$$

$$\mathbf{P} \bullet \mathbf{R} \rightarrow \begin{bmatrix} \mathbf{I} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \bullet \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} = [\mathbf{R}, \mathbf{p}] = \mathbf{F}$$

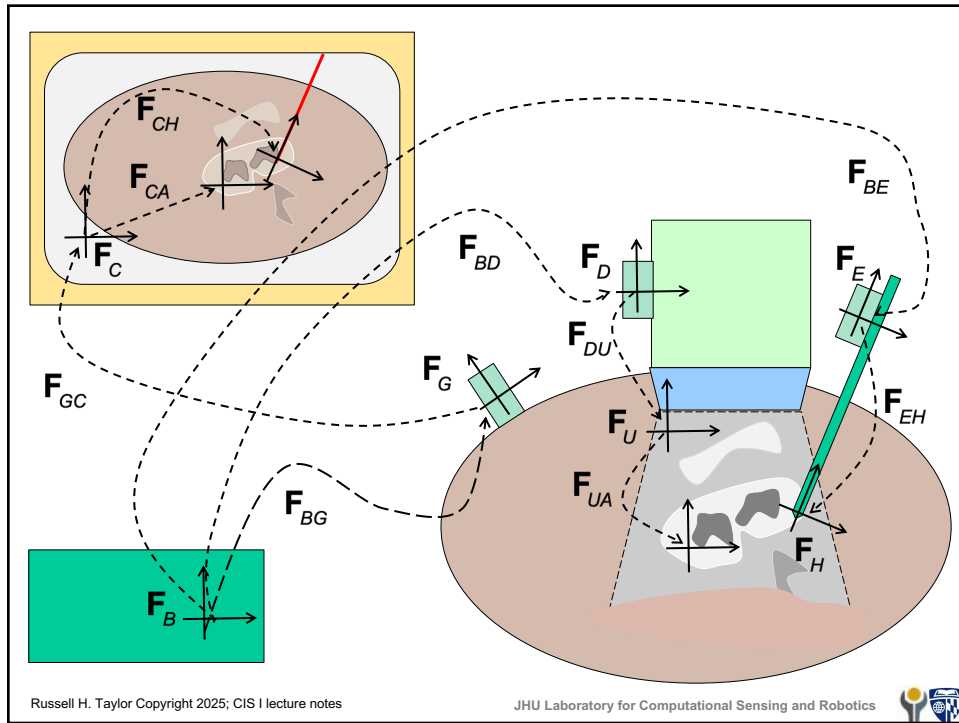
$$\mathbf{F} \bullet \mathbf{v} \rightarrow \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ 1 \end{bmatrix} = \begin{bmatrix} (\mathbf{R} \bullet \mathbf{v}) + \mathbf{p} \\ 1 \end{bmatrix}$$

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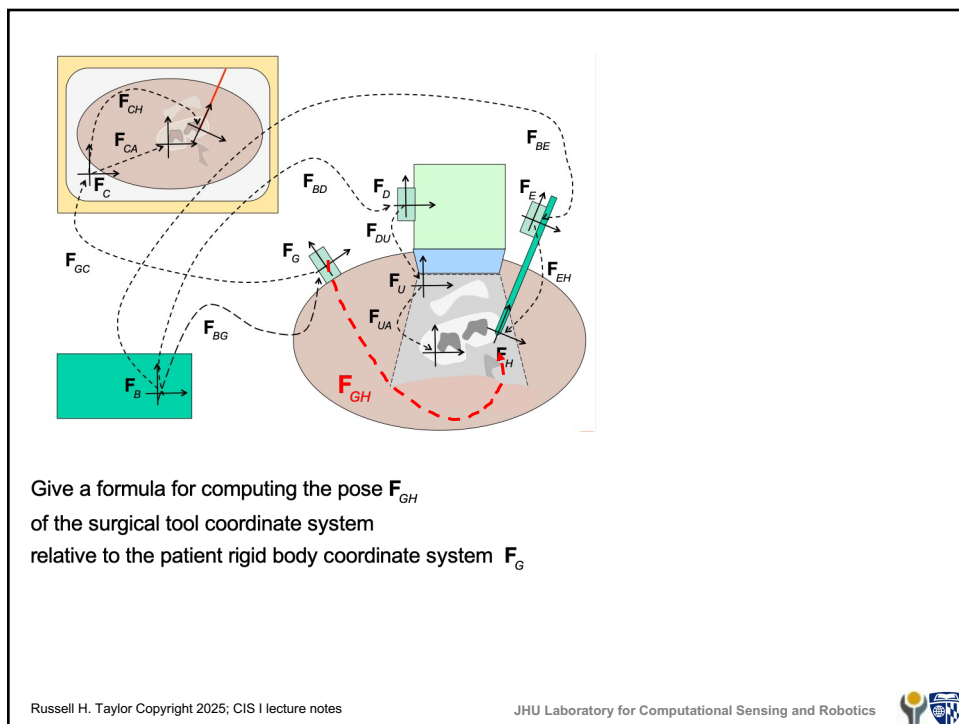
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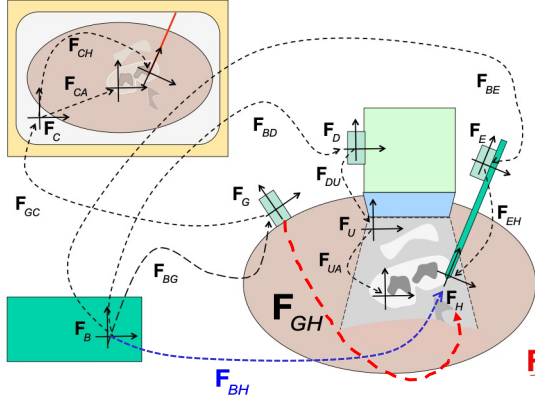
47



48



49



Give a formula for computing the pose F_{GH} of the surgical tool coordinate system relative to the patient rigid body coordinate system F_G ?

What are the components $F_{GH} = [R_{GH}, \vec{p}_{GH}]$?

$$F_{GH} = F_{BG}^{-1} F_{BE} F_{EH}$$

$$= F_{BG}^{-1} F_{BH}$$

$$R_{GH} = R_{BG}^{-1} R_{BE} R_{EH}$$

$$\vec{p}_{GH} = F_{BG}^{-1} \vec{p}_{BH} = F_{BG}^{-1} F_{BE} \vec{p}_{EH}$$

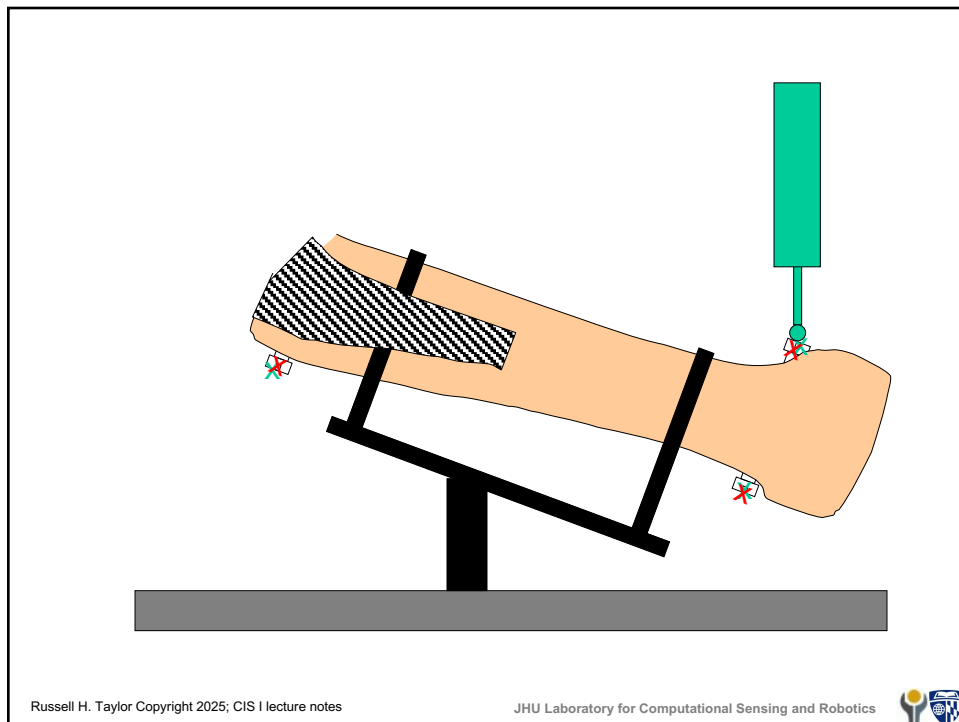
$$\vec{p}_{GH} = F_{BG}^{-1} (R_{BE} \vec{p}_{EH} + \vec{p}_{BE})$$

$$\vec{p}_{GH} = R_{BG}^{-1} (R_{BE} \vec{p}_{EH} + \vec{p}_{BE} - \vec{p}_{BG})$$

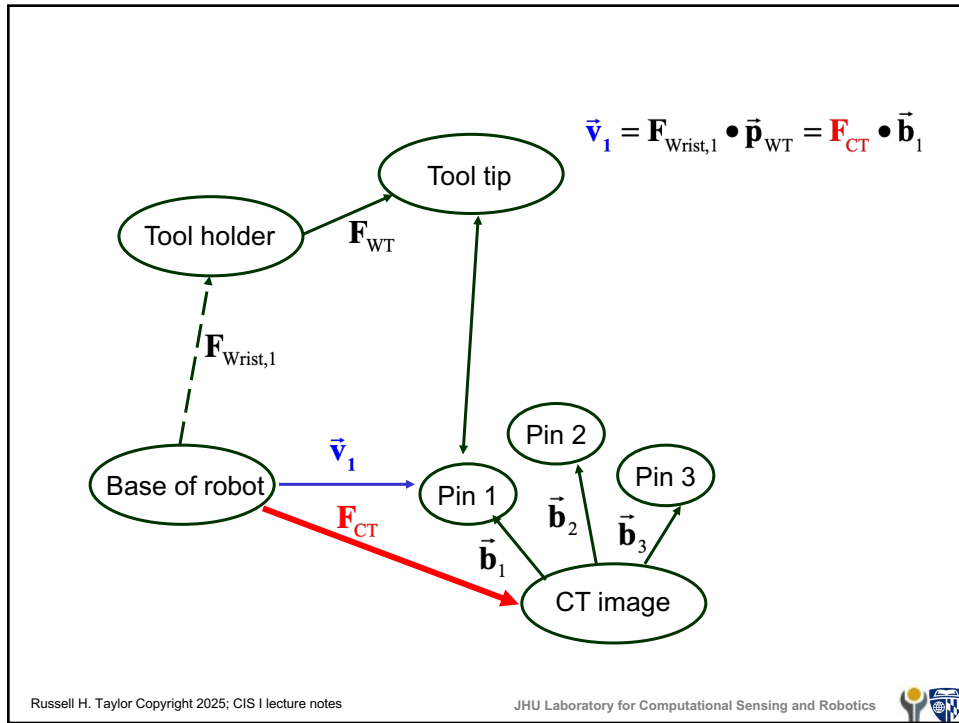
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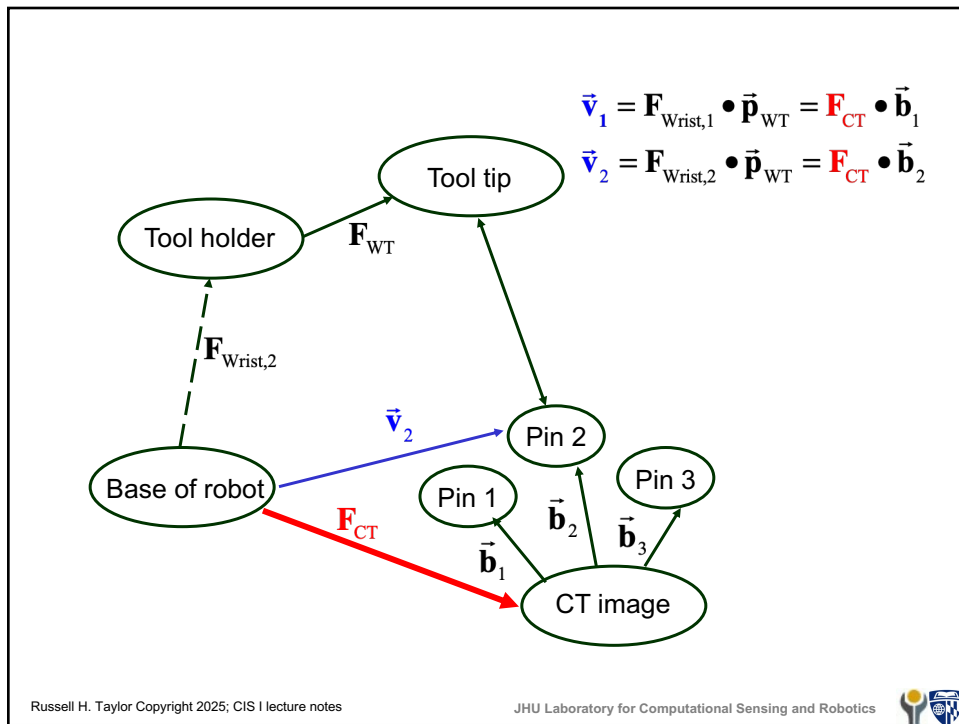
51



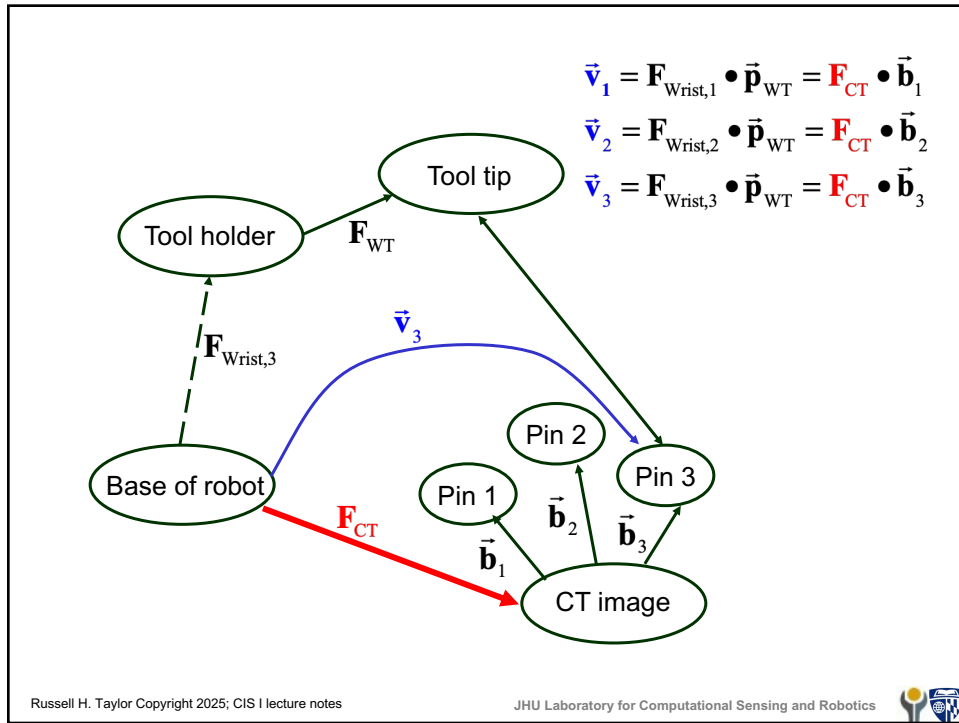
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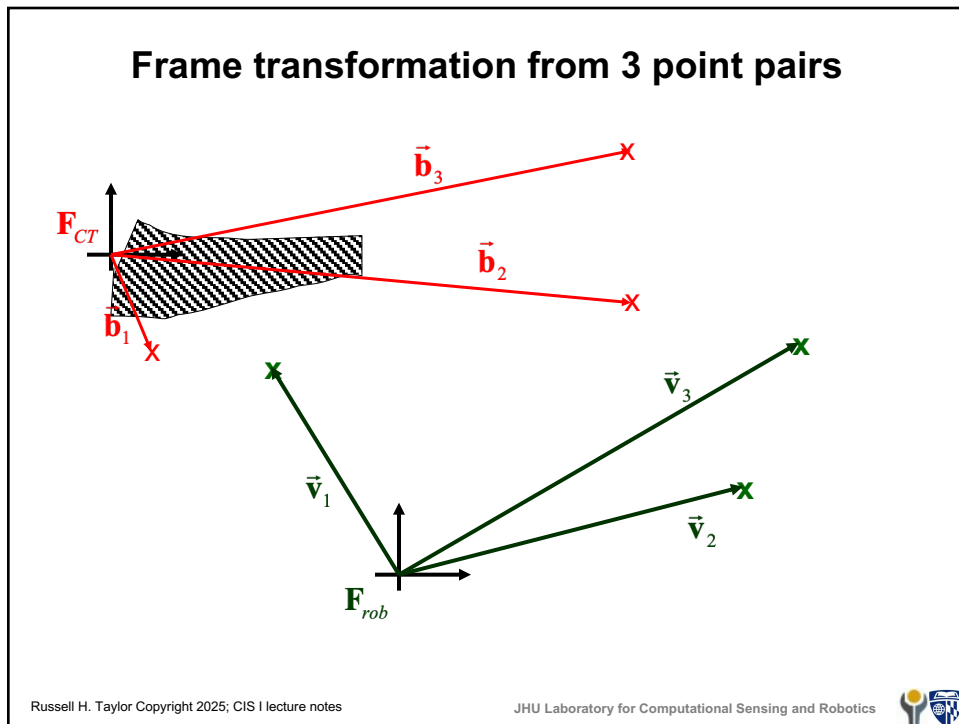
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55

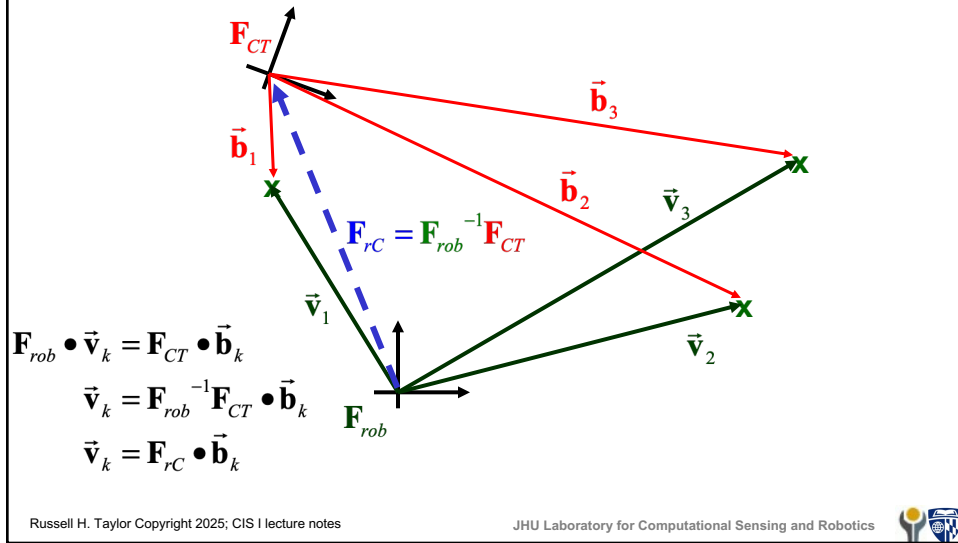


56



57

Frame transformation from 3 point pairs



58

Frame transformation from 3 point pairs

$$\vec{\mathbf{v}}_k = \mathbf{F}_{rC} \vec{\mathbf{b}}_k = \mathbf{R}_{rC} \vec{\mathbf{b}}_k + \vec{\mathbf{p}}_{rC}$$

Define

$$\vec{\mathbf{v}}_m = \frac{1}{3} \sum_1^3 \vec{\mathbf{v}}_k \quad \vec{\mathbf{b}}_m = \frac{1}{3} \sum_1^3 \vec{\mathbf{b}}_k$$

$$\vec{\mathbf{u}}_k = \vec{\mathbf{v}}_k - \vec{\mathbf{v}}_m \quad \vec{\mathbf{a}}_k = \vec{\mathbf{b}}_k - \vec{\mathbf{b}}_m$$

$$\mathbf{F}_{rC} \vec{\mathbf{a}}_k = \mathbf{R}_{rC} \vec{\mathbf{a}}_k + \vec{\mathbf{p}}_{rC}$$

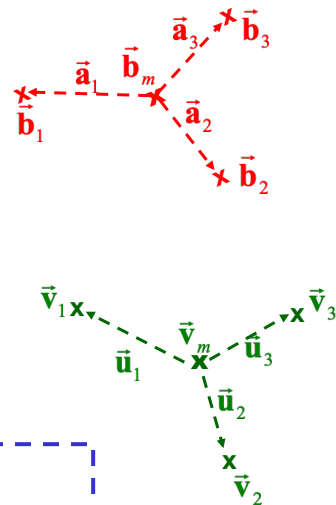
$$\mathbf{R}_{rC} \vec{\mathbf{a}}_k + \vec{\mathbf{p}}_{rC} = \mathbf{R}_{rC} (\vec{\mathbf{b}}_k - \vec{\mathbf{b}}_m) + \vec{\mathbf{p}}_{rC}$$

$$\mathbf{R}_{rC} \vec{\mathbf{a}}_k = \mathbf{R}_{rC} \vec{\mathbf{b}}_k + \vec{\mathbf{p}}_{rC} - \mathbf{R}_{rC} \vec{\mathbf{b}}_m - \vec{\mathbf{p}}_{rC}$$

$$\mathbf{R}_{rC} \vec{\mathbf{a}}_k = \vec{\mathbf{v}}_k - \vec{\mathbf{v}}_m = \vec{\mathbf{u}}_k$$

$$\vec{\mathbf{p}}_{rC} = \vec{\mathbf{v}}_m - \mathbf{R}_{rC} \vec{\mathbf{b}}_m$$

Solve These!!



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59

Rotation from multiple vector pairs

Given a system $\mathbf{R}\vec{\mathbf{a}}_k = \vec{\mathbf{u}}_k$ for $k = 1, \dots, n$ the problem is to estimate $\mathbf{R}$. This will require at least three such point pairs. Later in the course we will cover some good ways to solve this system. Here is a not-so-good way that will produce roughly correct answers:

Step 1: Form matrices $\mathbf{U} = \begin{bmatrix} \vec{\mathbf{u}}_1 & \dots & \vec{\mathbf{u}}_n \end{bmatrix}$ and $\mathbf{A} = \begin{bmatrix} \vec{\mathbf{a}}_1 & \dots & \vec{\mathbf{a}}_n \end{bmatrix}$

Step 2: Solve the system $\mathbf{R}\mathbf{A} = \mathbf{U}$ for $\mathbf{R}$. E.g., by $\mathbf{R} = \mathbf{U}\mathbf{A}^{-1}$

Step 3: Renormalize $\mathbf{R}$ to guarantee $\mathbf{R}^T\mathbf{R} = \mathbf{I}$.



Renormalizing Rotation Matrix

Given "rotation" matrix $\mathbf{R} = \begin{bmatrix} \vec{\mathbf{r}}_x & \vec{\mathbf{r}}_y & \vec{\mathbf{r}}_z \end{bmatrix}$, modify it so $\mathbf{R}^T\mathbf{R} = \mathbf{I}$.

Step 1: $\vec{\mathbf{a}} = \vec{\mathbf{r}}_y \times \vec{\mathbf{r}}_z$

Step 2: $\vec{\mathbf{b}} = \vec{\mathbf{r}}_z \times \vec{\mathbf{a}}$

Step 3: $\mathbf{R}_{normalized} = \begin{bmatrix} \frac{\vec{\mathbf{a}}}{\|\vec{\mathbf{a}}\|} & \frac{\vec{\mathbf{b}}}{\|\vec{\mathbf{b}}\|} & \frac{\vec{\mathbf{r}}_z}{\|\vec{\mathbf{r}}_z\|} \end{bmatrix}$



Iterative Improvement

Given a system $\mathbf{R}\bar{\mathbf{a}}_k = \bar{\mathbf{u}}_k$ for $k=1, \dots, n$ the problem is to estimate $\mathbf{R}$.

Step 0: $i \leftarrow 0$; $\mathbf{R}_0 \leftarrow$ initial estimate for $\mathbf{R}$

Step 2: Solve the system $\varepsilon = \arg \min_{\Delta \mathbf{R}_i} \sum_k \|\mathbf{R}_i \Delta \mathbf{R}_i \bar{\mathbf{a}}_k - \bar{\mathbf{u}}_k\|^2$

$$\varepsilon \approx \arg \min_{\theta} \sum_k \|\mathbf{R}_i \text{sk}(\vec{\theta}) \bar{\mathbf{a}}_k - \bar{\mathbf{u}}_k\|^2 = \arg \min_{\theta} \sum_k \|\mathbf{R}_i \text{sk}(\bar{\mathbf{a}}_k) \vec{\theta} + \bar{\mathbf{u}}_k\|^2$$

$\Delta \mathbf{R}_i = \text{Rot}(\vec{\theta})$ i.e., convert the angles into rotation

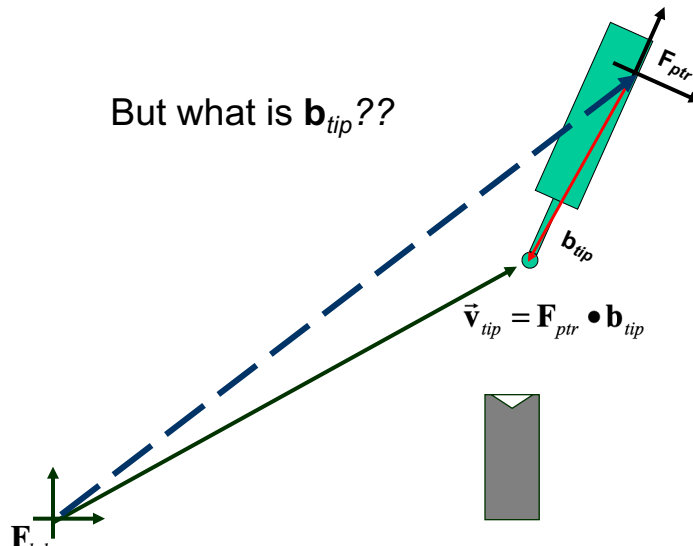
Step 3: $\mathbf{R}_{i+1} \leftarrow \mathbf{R}_i \Delta \mathbf{R}_i$; $i \leftarrow i+1$

Step 4: if ε is small enough, then done; else go to step 2



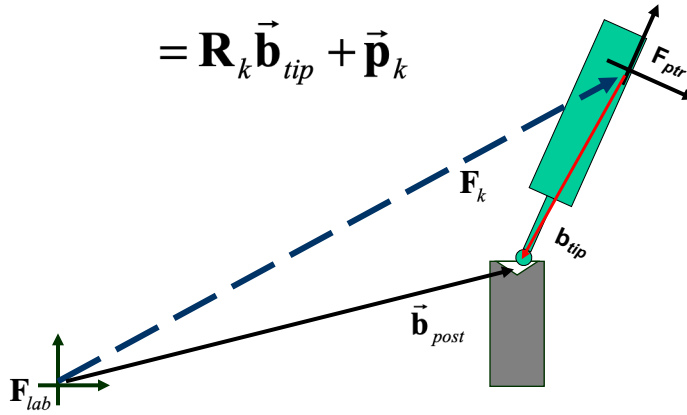
Calibrating a pointer

But what is $\mathbf{b}_{tip}$??



Calibrating a pointer

$$\begin{aligned}\vec{\mathbf{b}}_{post} &= \mathbf{F}_k \vec{\mathbf{b}}_{tip} \\ &= \mathbf{R}_k \vec{\mathbf{b}}_{tip} + \vec{\mathbf{p}}_k\end{aligned}$$



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64

Calibrating a pointer

For each measurement k , we have

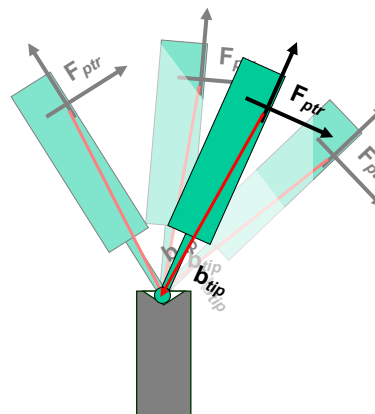
$$\vec{\mathbf{b}}_{post} = \mathbf{R}_k \vec{\mathbf{b}}_{tip} + \vec{\mathbf{p}}_k$$

i. e.,

$$\mathbf{R}_k \vec{\mathbf{b}}_{tip} - \vec{\mathbf{b}}_{post} = -\vec{\mathbf{p}}_k$$

Set up a least squares problem

$$\begin{bmatrix} \vdots & \vdots \\ \mathbf{R}_k & -\mathbf{I} \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} \vec{\mathbf{b}}_{tip} \\ \vec{\mathbf{b}}_{post} \end{bmatrix} \cong \begin{bmatrix} \vdots \\ -\vec{\mathbf{p}}_k \\ \vdots \end{bmatrix}$$



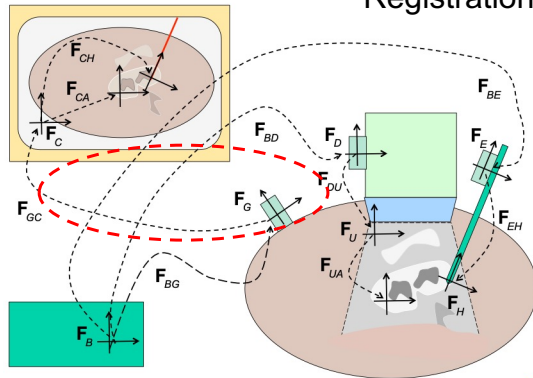
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65

“Registration Transformations”



Given a coordinate system F_C and another coordinate system F_G (e.g., a CT scan and a tracked "rigid body" attached to the patient, and points $\vec{c}_i$ in the coordinate system F_C and points $\vec{g}_i$ in the coordinate system F_G , then the "registration transformation" F_{GC} between F_G and F_C one in which $F_{GC} \vec{c}_i = \vec{g}_i$ if and only if $\vec{c}_i$ and $\vec{g}_i$ refer to the same or corresponding points.

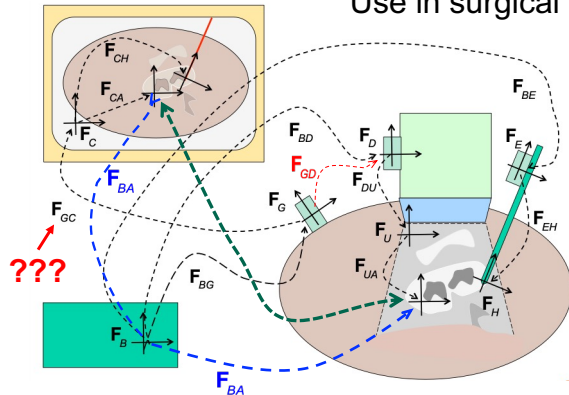
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66

Use in surgical navigation



$$\begin{aligned}
 F_{BA} &= F_{BD} F_{DU} F_{UA} \\
 F_{BA} &= F_{BG} F_{GC} F_{CA} \\
 F_{BG} F_{GC} F_{CA} &= F_{BD} F_{DU} F_{UA} \\
 F_{CA} &= (F_{BG} F_{GC})^{-1} F_{BD} F_{DU} F_{UA} \\
 F_{GC} &= F_{BG}^{-1} F_{BD} F_{DU} F_{UA} F_{CA}^{-1} \\
 F_{GC} &= F_{GD} F_{DU} F_{UA} F_{CA}^{-1} \\
 \text{where } F_{GD} &= F_{BG}^{-1} F_{BD}
 \end{aligned}$$

If an anatomic structure is identified at pose F_{UA} in ultrasound image coordinates give the formula for computing the corresponding pose F_{CA} in CT coordinates

NOTE: In cases where there is a reference body (here F_G) affixed to the patient, it is sometimes convenient to assume that the navigation software routinely transforms all tracked markers to the coordinate system of F_G (i.e., $F_{GX} = F_{BG}^{-1} F_{BX}$).

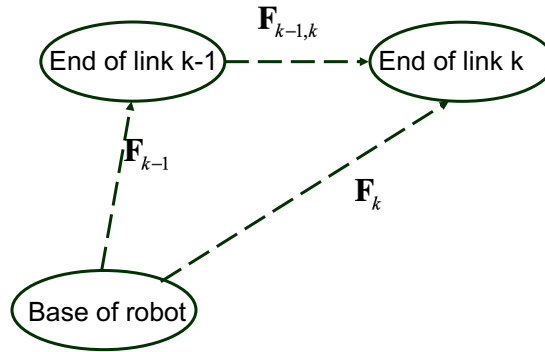
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67

Kinematic Links



$$\mathbf{F}_k = \mathbf{F}_{k-1} \bullet \mathbf{F}_{k-1,k}$$

$$[\mathbf{R}_k, \vec{\mathbf{p}}_k] = [\mathbf{R}_{k-1}, \mathbf{p}_{k-1}] \bullet [\mathbf{R}_{k-1,k}, \mathbf{p}_{k-1,k}]$$

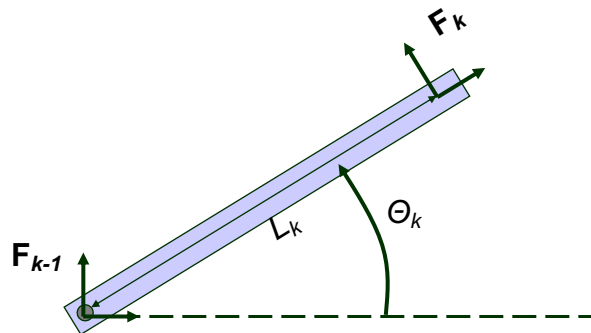
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68

Kinematic Links



$$\mathbf{F}_k = \mathbf{F}_{k-1} \bullet \mathbf{F}_{k-1,k}$$

$$[\mathbf{R}_k, \vec{\mathbf{p}}_k] = [\mathbf{R}_{k-1}, \mathbf{p}_{k-1}] \bullet [\mathbf{R}_{k-1,k}, \mathbf{p}_{k-1,k}]$$

$$= [\mathbf{R}_{k-1}, \mathbf{p}_{k-1}] \bullet [\text{Rot}(\vec{\mathbf{r}}_k, \theta_k), \vec{\mathbf{0}}] \bullet [\mathbf{I}, L_k \vec{\mathbf{x}}]$$

$$= [\mathbf{R}_{k-1}, \mathbf{p}_{k-1}] \bullet [\text{Rot}(\vec{\mathbf{r}}_k, \theta_k), L_k \text{Rot}(\vec{\mathbf{r}}_k, \theta_k) \bullet \vec{\mathbf{x}}]$$

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69

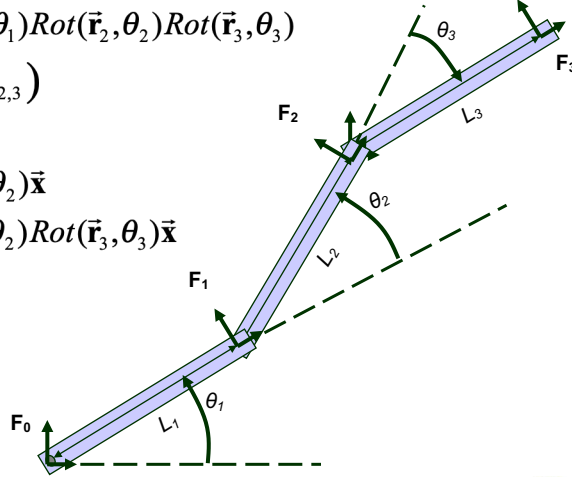
Kinematic Chains

$$\mathbf{F}_0 = [\mathbf{I}, \vec{0}]$$

$$\mathbf{R}_3 = \mathbf{R}_{0,1} \mathbf{R}_{1,2} \mathbf{R}_{2,3} = \text{Rot}(\vec{\mathbf{r}}_1, \theta_1) \text{Rot}(\vec{\mathbf{r}}_2, \theta_2) \text{Rot}(\vec{\mathbf{r}}_3, \theta_3)$$

$$\vec{\mathbf{p}}_3 = \vec{\mathbf{p}}_{0,1} + \mathbf{R}_{0,1} (\vec{\mathbf{p}}_{1,2} + \mathbf{R}_{1,2} \vec{\mathbf{p}}_{2,3})$$

$$\begin{aligned} &= L_1 \text{Rot}(\vec{\mathbf{r}}_1, \theta_1) \vec{\mathbf{x}} \\ &\quad + L_2 \text{Rot}(\vec{\mathbf{r}}_1, \theta_1) \text{Rot}(\vec{\mathbf{r}}_2, \theta_2) \vec{\mathbf{x}} \\ &\quad + L_3 \text{Rot}(\vec{\mathbf{r}}_1, \theta_1) \text{Rot}(\vec{\mathbf{r}}_2, \theta_2) \text{Rot}(\vec{\mathbf{r}}_3, \theta_3) \vec{\mathbf{x}} \end{aligned}$$



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70

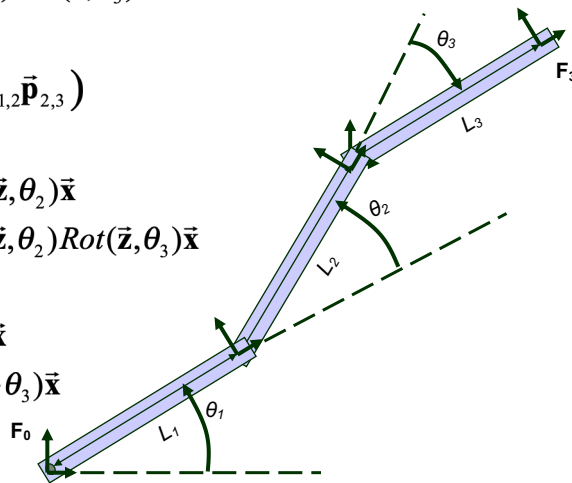
Kinematic Chains

$$\text{If } \vec{\mathbf{r}}_1 = \vec{\mathbf{r}}_2 = \vec{\mathbf{r}}_3 = \vec{\mathbf{z}},$$

$$\begin{aligned} \mathbf{R}_3 &= \text{Rot}(\vec{\mathbf{z}}, \theta_1) \text{Rot}(\vec{\mathbf{z}}, \theta_2) \text{Rot}(\vec{\mathbf{z}}, \theta_3) \\ &= \text{Rot}(\vec{\mathbf{z}}, \theta_1 + \theta_2 + \theta_3) \end{aligned}$$

$$\vec{\mathbf{p}}_3 = \vec{\mathbf{p}}_{0,1} + \mathbf{R}_{0,1} (\vec{\mathbf{p}}_{1,2} + \mathbf{R}_{1,2} \vec{\mathbf{p}}_{2,3})$$

$$\begin{aligned} &= L_1 \text{Rot}(\vec{\mathbf{z}}, \theta_1) \vec{\mathbf{x}} \\ &\quad + L_2 \text{Rot}(\vec{\mathbf{z}}, \theta_1) \text{Rot}(\vec{\mathbf{z}}, \theta_2) \vec{\mathbf{x}} \\ &\quad + L_3 \text{Rot}(\vec{\mathbf{z}}, \theta_1) \text{Rot}(\vec{\mathbf{z}}, \theta_2) \text{Rot}(\vec{\mathbf{z}}, \theta_3) \vec{\mathbf{x}} \\ &= L_1 \text{Rot}(\vec{\mathbf{z}}, \theta_1) \vec{\mathbf{x}} \\ &\quad + L_2 \text{Rot}(\vec{\mathbf{z}}, \theta_1 + \theta_2) \vec{\mathbf{x}} \\ &\quad + L_3 \text{Rot}(\vec{\mathbf{z}}, \theta_1 + \theta_2 + \theta_3) \vec{\mathbf{x}} \end{aligned}$$



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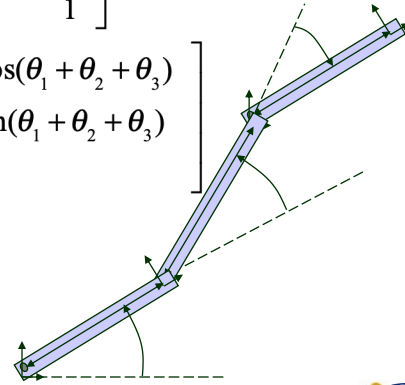
71

Kinematic Chains

If $\vec{r}_1 = \vec{r}_2 = \vec{r}_3 = \vec{z}$,

$$\mathbf{R}_3 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\vec{p}_3 = \begin{bmatrix} L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3) \\ 0 \end{bmatrix}$$



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72

“Small” Transformations

- A great deal of CIS is concerned with computing and using geometric information based on imprecise knowledge
- Similarly, one is often concerned with the effects of relatively small rotations and displacements
- Essentially, we will be using fairly straightforward linearizations to model these situations, but a specialized notation is often useful

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74

“Small” Frame Transformations

Represent a "small" pose shift consisting of a small rotation $\Delta\mathbf{R}$ followed by a small displacement $\Delta\vec{\mathbf{p}}$ as

$$\Delta\mathbf{F} = [\Delta\mathbf{R}, \Delta\vec{\mathbf{p}}]$$

Then

$$\Delta\mathbf{F} \bullet \vec{\mathbf{v}} = \Delta\mathbf{R} \bullet \vec{\mathbf{v}} + \Delta\vec{\mathbf{p}}$$



Small Rotations

$\Delta\mathbf{R}$ = a small rotation

$\mathbf{R}_{\vec{\mathbf{a}}}(\Delta\alpha)$ = a rotation by a small angle $\Delta\alpha$ about axis $\vec{\mathbf{a}}$

$\text{Rot}(\vec{\mathbf{a}}, \|\vec{\mathbf{a}}\|) \bullet \vec{\mathbf{b}} \approx \vec{\mathbf{a}} \times \vec{\mathbf{b}} + \vec{\mathbf{b}}$ for $\|\vec{\mathbf{a}}\|$ sufficiently small

$\Delta\mathbf{R}(\vec{\mathbf{a}})$ = a rotation that is small enough so that any error introduced by this approximation is negligible

$$\Delta\mathbf{R}(\lambda \vec{\mathbf{a}}) \bullet \Delta\mathbf{R}(\mu \vec{\mathbf{b}}) \cong \Delta\mathbf{R}(\lambda \vec{\mathbf{a}} + \mu \vec{\mathbf{b}}) \quad (\text{Linearity for small rotations})$$

Exercise: Work out the linearity proposition by substitution



Approximations to “Small” Frames

$$\begin{aligned}\Delta\mathbf{F}(\vec{\mathbf{a}}, \Delta\vec{\mathbf{p}}) &\triangleq [\Delta\mathbf{R}(\vec{\mathbf{a}}), \Delta\vec{\mathbf{p}}] \\ \Delta\mathbf{F}(\vec{\mathbf{a}}, \Delta\vec{\mathbf{p}}) \bullet \vec{\mathbf{v}} &= \Delta\mathbf{R}(\vec{\mathbf{a}}) \bullet \vec{\mathbf{v}} + \Delta\vec{\mathbf{p}} \\ &\approx \vec{\mathbf{v}} + \vec{\mathbf{a}} \times \vec{\mathbf{v}} + \Delta\vec{\mathbf{p}}\end{aligned}$$

$$\begin{aligned}\vec{\mathbf{a}} \times \vec{\mathbf{v}} &= \text{skew}(\vec{\mathbf{a}}) \bullet \vec{\mathbf{v}} \\ &= \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \bullet \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \\ \text{skew}(\vec{\mathbf{a}}) \bullet \vec{\mathbf{a}} &= \vec{\mathbf{a}} \times \vec{\mathbf{a}} = \vec{\mathbf{0}}\end{aligned}$$

$$\begin{aligned}\Delta\mathbf{R}(\vec{\mathbf{a}}) &\approx \mathbf{I} + \text{skew}(\vec{\mathbf{a}}) \\ \Delta\mathbf{R}(\vec{\mathbf{a}})^{-1} &\approx \mathbf{I} - \text{skew}(\vec{\mathbf{a}}) = \mathbf{I} + \text{skew}(-\vec{\mathbf{a}})\end{aligned}$$



Approximations to “Small” Frames

Notational NOTE:

We often use $\vec{\alpha}$ to represent a vector of small angles
and $\vec{\epsilon}$ to represent a vector of small displacements

In using these approximations, we typically ignore second order terms. I.e.,

$$\vec{\alpha}_A \vec{\alpha}_B \approx \vec{0}, \vec{\alpha}_A \vec{\epsilon}_B \approx \vec{0}, \vec{\epsilon}_A \vec{\epsilon}_B \approx \vec{0}, \text{ etc.}$$



Errors & sensitivity

Often, we do not have an accurate value for a transformation, so we need to model the error. We model this as a composition of a "nominal" frame and a small displacement

$$\mathbf{F}_{\text{actual}} = \mathbf{F}_{\text{nominal}} \bullet \Delta \mathbf{F}$$

Often, we will use the notation $\mathbf{F}^*$ for $\mathbf{F}_{\text{actual}}$ and will just use $\mathbf{F}$ for $\mathbf{F}_{\text{nominal}}$. Thus we may write something like

$$\mathbf{F}^* = \mathbf{F} \bullet \Delta \mathbf{F}$$

or (less often) $\mathbf{F}^* = \Delta \mathbf{F} \bullet \mathbf{F}$. We also use $\vec{\mathbf{v}}^* = \vec{\mathbf{v}} + \Delta \vec{\mathbf{v}}$, etc.

Thus, if we use the former form (error on the right), and

have nominal relationship $\vec{\mathbf{v}} = \mathbf{F} \bullet \vec{\mathbf{b}}$, we get

$$\begin{aligned} \vec{\mathbf{v}}^* &= \mathbf{F}^* \bullet \vec{\mathbf{b}}^* \\ &= \mathbf{F} \bullet \Delta \mathbf{F} \bullet (\vec{\mathbf{b}} + \Delta \vec{\mathbf{b}}) = \mathbf{F} \bullet (\Delta \mathbf{R} \bullet \vec{\mathbf{b}} + \Delta \mathbf{R} \bullet \Delta \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) \\ &\approx \mathbf{R} \bullet ((\mathbf{I} + \text{sk}(\vec{\alpha})) \bullet (\vec{\mathbf{b}} + \Delta \vec{\mathbf{b}}) + \Delta \vec{\mathbf{p}}) + \vec{\mathbf{p}} = \mathbf{R} \bullet (\vec{\mathbf{b}} + \vec{\alpha} \times \vec{\mathbf{b}} + \Delta \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) + \vec{\mathbf{p}} \\ &\approx \mathbf{R} \bullet (\vec{\alpha} \times \vec{\mathbf{b}} + \Delta \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) + \mathbf{R} \bullet \vec{\mathbf{b}} + \vec{\mathbf{p}} = \mathbf{R} \bullet (\vec{\alpha} \times \vec{\mathbf{b}} + \Delta \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) + \vec{\mathbf{v}} \\ \Delta \vec{\mathbf{v}} &\approx \mathbf{R} \bullet (\vec{\alpha} \times \vec{\mathbf{b}} + \Delta \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) \end{aligned}$$

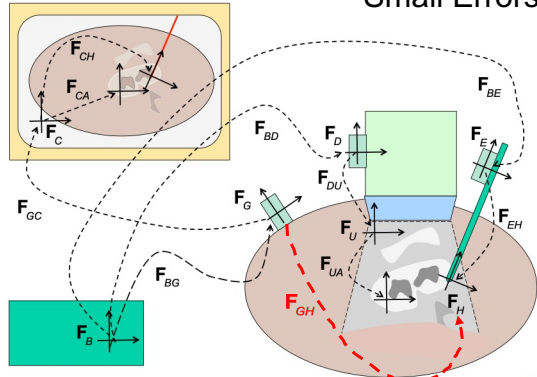
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79

"Small Errors"



Suppose that there is a small systematic error in the tracking system so that

$$\mathbf{F}_{Bx}^* = \Delta \mathbf{F}_B \mathbf{F}_{Bx}$$

for $\mathbf{F}_{BG}$, $\mathbf{F}_{BD}$, $\mathbf{F}_{BE}$. How does this affect the calculation of $\mathbf{F}_{GH}$?

$$\begin{aligned} \mathbf{F}_{GH}^* &= (\mathbf{F}_{BG}^*)^{-1} \mathbf{F}_{BE}^* \mathbf{F}_{EH} \\ \mathbf{F}_{GH} \Delta \mathbf{F}_{GH} &= (\Delta \mathbf{F}_B \mathbf{F}_{BG})^{-1} \Delta \mathbf{F}_B \mathbf{F}_{BE} \mathbf{F}_{EH} \\ \Delta \mathbf{F}_{GH} &= \mathbf{F}_{GH}^{-1} (\Delta \mathbf{F}_B \mathbf{F}_{BG})^{-1} \Delta \mathbf{F}_B \mathbf{F}_{BE} \mathbf{F}_{EH} \\ &= \mathbf{F}_{GH}^{-1} \mathbf{F}_{BG}^{-1} \Delta \mathbf{F}_B^{-1} \Delta \mathbf{F}_B \mathbf{F}_{BE} \mathbf{F}_{EH} \\ &= \mathbf{F}_{GH}^{-1} \mathbf{F}_{BG}^{-1} \mathbf{F}_{BE} \mathbf{F}_{EH} = \mathbf{F}_{GH}^{-1} \mathbf{F}_{GH} = \mathbf{I} \end{aligned}$$

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80

"Small Errors"

Suppose that there are additional errors in the tracking of each tracker body so that

$$\mathbf{F}_{Bx}^* = \Delta \mathbf{F}_B \mathbf{F}_{Bx} \Delta \mathbf{F}_{Bx}$$

for $\mathbf{F}_{BG}^*$, $\mathbf{F}_{BD}^*$, $\mathbf{F}_{BE}^*$. How does this affect the calculation of $\mathbf{F}_{GH}^*$?

$$\begin{aligned} \mathbf{F}_{GH}^* &= \mathbf{F}_{GH} \Delta \mathbf{F}_{GH} = (\mathbf{F}_{BG}^*)^{-1} \mathbf{F}_{BE}^* \mathbf{F}_{EH} \\ \Delta \mathbf{F}_{GH} &= \mathbf{F}_{GH}^{-1} (\Delta \mathbf{F}_B \mathbf{F}_{BG} \Delta \mathbf{F}_{BG})^{-1} (\Delta \mathbf{F}_B \mathbf{F}_{BE} \Delta \mathbf{F}_{BE}) \mathbf{F}_{EH} \\ \Delta \mathbf{F}_{GH} &= (\mathbf{F}_{BG}^{-1} \mathbf{F}_{BE} \mathbf{F}_{EH})^{-1} \Delta \mathbf{F}_{BG}^{-1} \mathbf{F}_{BG}^{-1} \Delta \mathbf{F}_B^{-1} \Delta \mathbf{F}_B \mathbf{F}_{BE} \Delta \mathbf{F}_{BE} \mathbf{F}_{EH} \\ &= \mathbf{F}_{EH}^{-1} \mathbf{F}_{BE}^{-1} \mathbf{F}_{BG} \Delta \mathbf{F}_{BG}^{-1} \mathbf{F}_{BG}^{-1} \mathbf{F}_{BE} \Delta \mathbf{F}_{BE} \mathbf{F}_{EH} \end{aligned}$$

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81

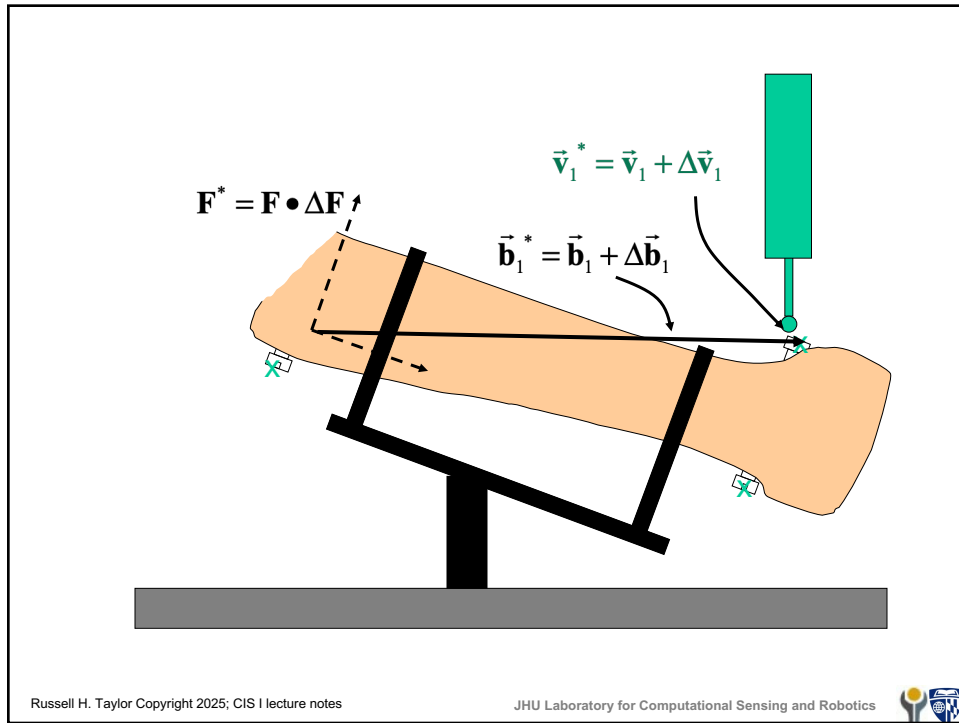
$\mathbf{F} = [\mathbf{R}, \mathbf{p}]$

$\vec{\mathbf{b}}_1$

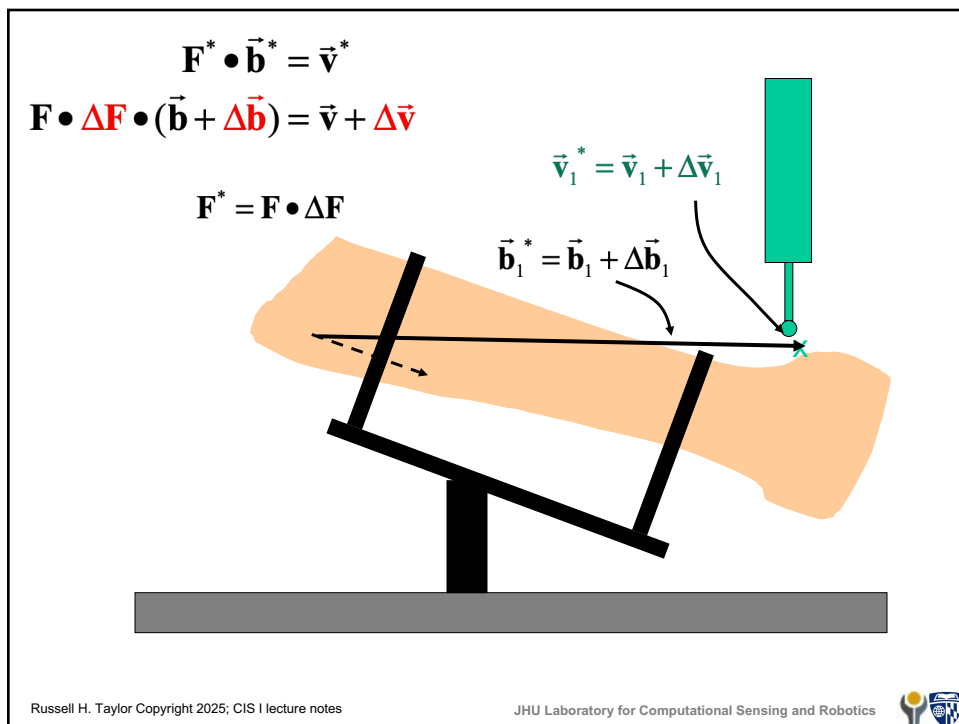
$\vec{\mathbf{v}}_1$

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82



83



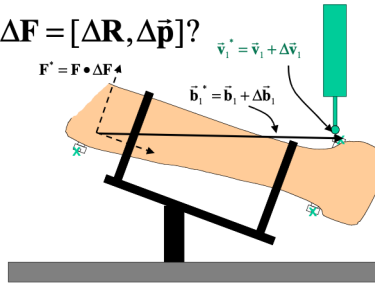
84

Errors & Sensitivity

Suppose that we know nominal values for $\mathbf{F}$, $\vec{\mathbf{b}}$, and $\vec{\mathbf{v}}$ and that

$$\begin{bmatrix} -\varepsilon, -\varepsilon, -\varepsilon \end{bmatrix}^T \leq \Delta \vec{\mathbf{v}}_1 \leq \begin{bmatrix} \varepsilon, \varepsilon, \varepsilon \end{bmatrix}^T \quad (\text{i.e., } \|\Delta \vec{\mathbf{v}}_1\|_\infty \leq \varepsilon)$$

What does this tell us about $\Delta \mathbf{F} = [\Delta \mathbf{R}, \Delta \vec{\mathbf{p}}]$?



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85

Errors & Sensitivity

$$\begin{aligned} \vec{\mathbf{v}}^* &= \mathbf{F}^* \bullet \vec{\mathbf{b}}^* \\ &= \mathbf{F} \bullet \Delta \mathbf{F} \bullet (\vec{\mathbf{b}} + \Delta \vec{\mathbf{b}}) \\ &= \mathbf{R} \bullet (\Delta \mathbf{R}(\vec{\alpha}) \bullet (\vec{\mathbf{b}} + \Delta \vec{\mathbf{b}}) + \Delta \vec{\mathbf{p}}) + \vec{\mathbf{p}} \\ &\approx \mathbf{R} \bullet (\vec{\mathbf{b}} + \Delta \vec{\mathbf{b}} + \vec{\alpha} \times \vec{\mathbf{b}} + \vec{\alpha} \times \Delta \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) + \vec{\mathbf{p}} \\ &= \mathbf{R} \bullet \vec{\mathbf{b}} + \vec{\mathbf{p}} + \mathbf{R} \bullet (\Delta \vec{\mathbf{b}} + \vec{\alpha} \times \vec{\mathbf{b}} + \vec{\alpha} \times \Delta \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) \\ &\approx \vec{\mathbf{v}} + \mathbf{R} \bullet (\Delta \vec{\mathbf{b}} + \vec{\alpha} \times \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) \end{aligned}$$

if $\|\vec{\alpha} \times \Delta \vec{\mathbf{b}}\| \leq \|\vec{\alpha}\| \|\Delta \vec{\mathbf{b}}\|$ is negligible (it usually is)

so

$$\Delta \vec{\mathbf{v}} = \vec{\mathbf{v}}^* - \vec{\mathbf{v}} \approx \mathbf{R} \bullet (\Delta \vec{\mathbf{b}} + \vec{\alpha} \times \vec{\mathbf{b}} + \Delta \vec{\mathbf{p}}) = \mathbf{R} \bullet \Delta \vec{\mathbf{b}} + \mathbf{R} \bullet \vec{\alpha} \times \vec{\mathbf{b}} + \mathbf{R} \bullet \Delta \vec{\mathbf{p}}$$

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86

Digression: “rotation triple product”

Expressions like $\mathbf{R} \bullet \vec{\mathbf{a}} \times \vec{\mathbf{b}}$ are linear in $\vec{\mathbf{a}}$, but are not always convenient to work with. Often we would prefer something like $\mathbf{M}(\mathbf{R}, \vec{\mathbf{b}}) \bullet \vec{\mathbf{a}}$.

$$\begin{aligned}\mathbf{R} \bullet \vec{\mathbf{a}} \times \vec{\mathbf{b}} &= -\mathbf{R} \bullet \vec{\mathbf{b}} \times \vec{\mathbf{a}} \\ &= \mathbf{R} \bullet \text{skew}(-\vec{\mathbf{b}}) \bullet \vec{\mathbf{a}} \\ &= \left[\mathbf{R} \bullet \text{skew}(\vec{\mathbf{b}})^T \right] \bullet \vec{\mathbf{a}}\end{aligned}$$



Digression: “rotation triple product”

Here are a few more useful facts:

$$\begin{aligned}\mathbf{R} \bullet (\vec{\mathbf{a}} \times \vec{\mathbf{b}}) &= (\mathbf{R} \bullet \vec{\mathbf{a}}) \times (\mathbf{R} \bullet \vec{\mathbf{b}}) \\ \vec{\mathbf{a}} \times (\mathbf{R} \bullet \vec{\mathbf{b}}) &= \mathbf{R} \bullet ((\mathbf{R}^{-1} \bullet \vec{\mathbf{a}}) \times \vec{\mathbf{b}})\end{aligned}$$

Consequently

$$\begin{aligned}\text{skew}(\vec{\mathbf{a}}) \bullet \mathbf{R} &= \mathbf{R} \bullet \text{skew}(\mathbf{R}^{-1} \bullet \vec{\mathbf{a}}) \\ \mathbf{R}^{-1} \text{skew}(\vec{\mathbf{a}}) \bullet \mathbf{R} &= \text{skew}(\mathbf{R}^{-1} \bullet \vec{\mathbf{a}}) \\ \mathbf{R} \bullet \text{skew}(\vec{\mathbf{a}}) \bullet \mathbf{R}^{-1} &= \text{skew}(\mathbf{R} \bullet \vec{\mathbf{a}})\end{aligned}$$



A “standard form” for linearized error expressions

It is often convenient to use identities to rearrange expressions involving small error variables into sums of terms with the general form $\mathbf{M}_k \vec{\eta}_k$, where $\mathbf{M}_k$ involve things known to the computer, and the $\vec{\eta}_k$ are error variables.

For example,

$$\vec{\gamma} = \mathbf{R} \text{sk}(\vec{\alpha}) \vec{a} + \text{sk}(\vec{\beta}) \vec{b}$$

would be rewritten as

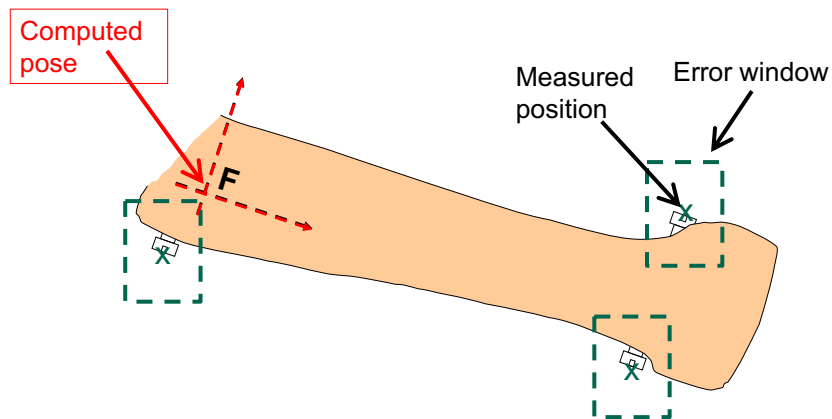
$$\vec{\gamma} = -\mathbf{R} \text{sk}(\vec{a}) \vec{\alpha} - \text{sk}(\vec{b}) \vec{\beta}$$

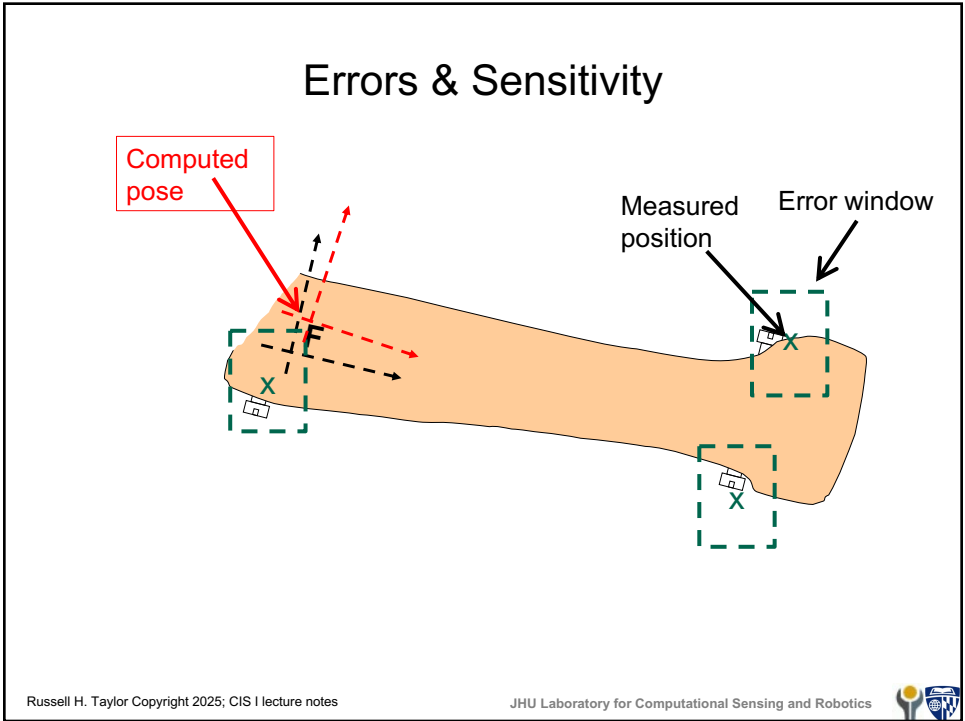
or

$$\vec{\gamma} = \mathbf{R} \text{sk}(-\vec{a}) \vec{\alpha} + \text{sk}(-\vec{b}) \vec{\beta}$$

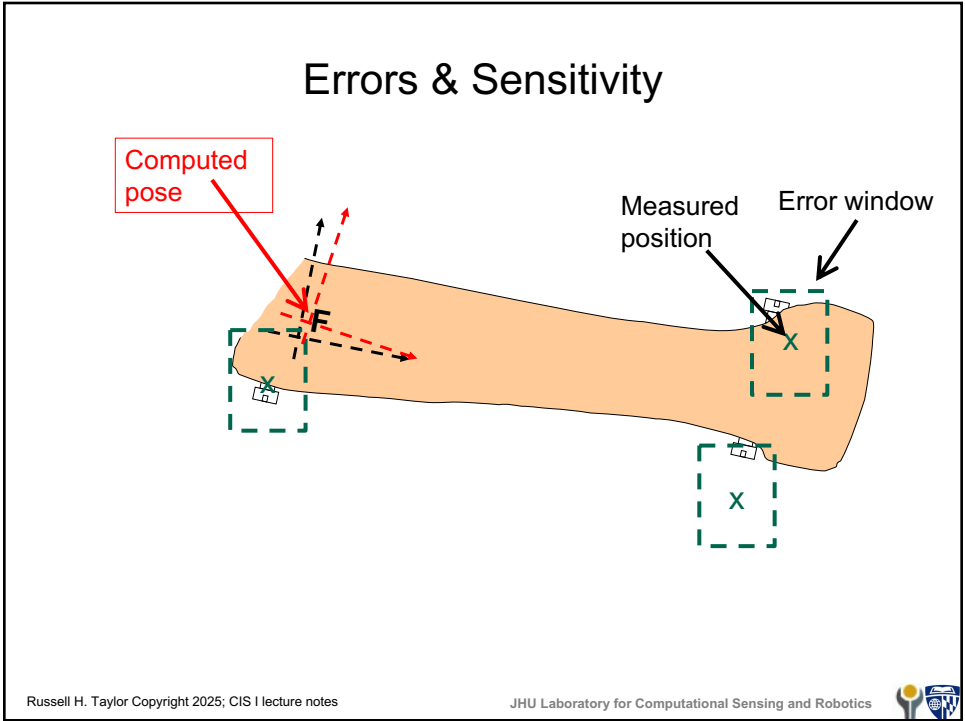


Errors & Sensitivity

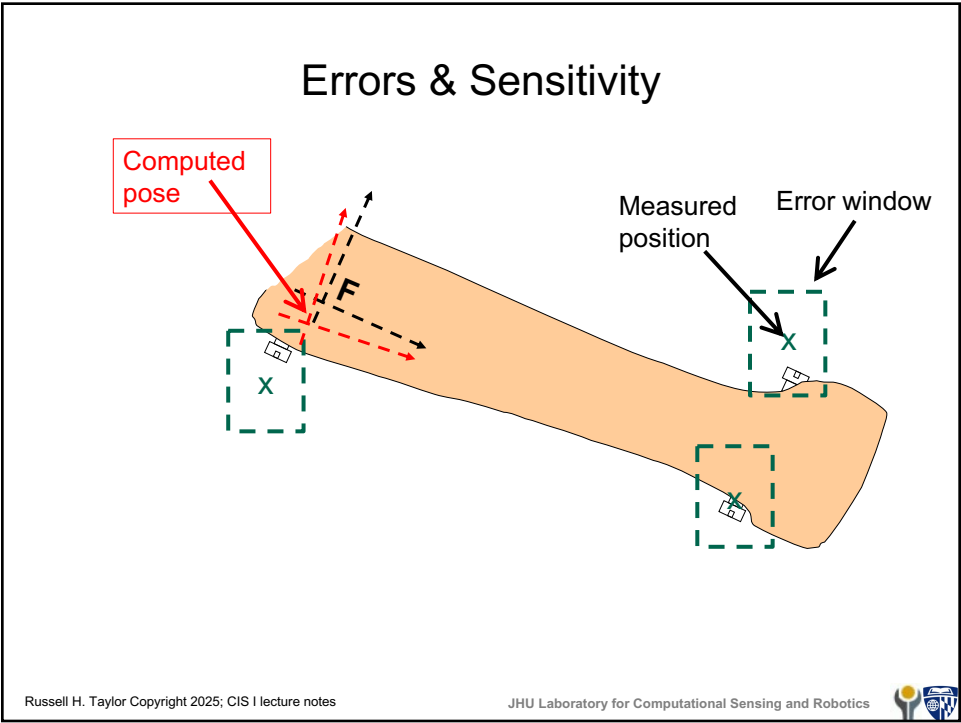




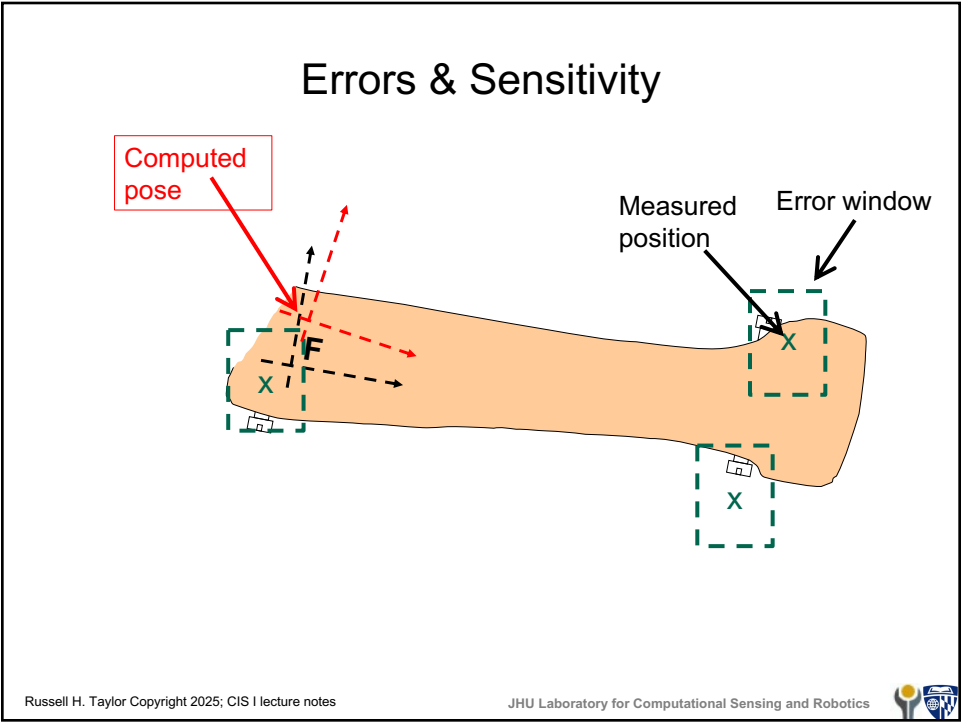
91



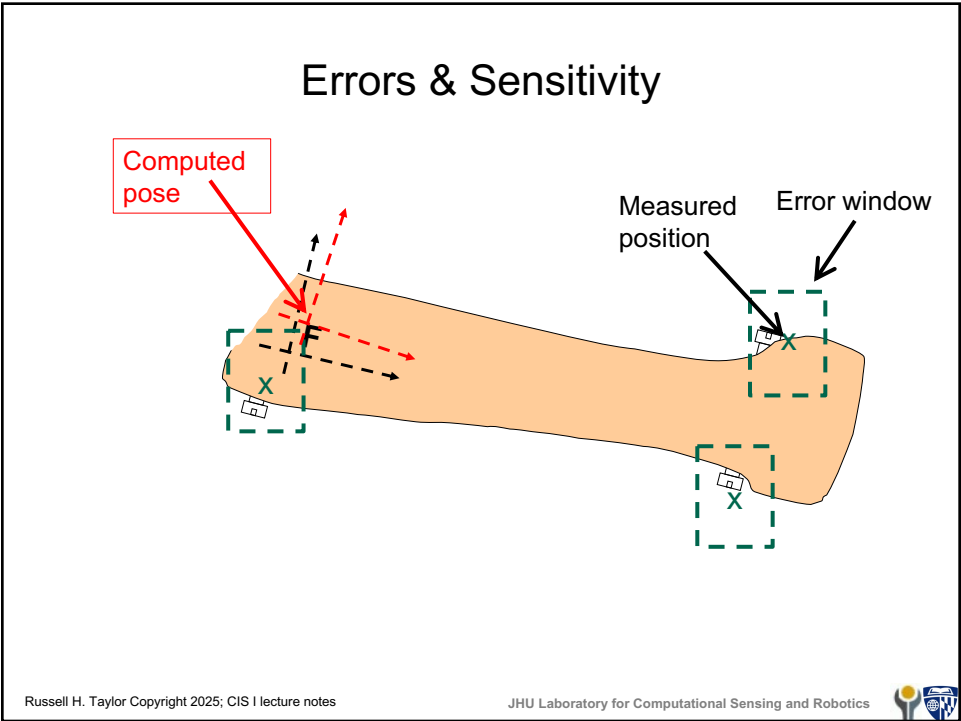
92



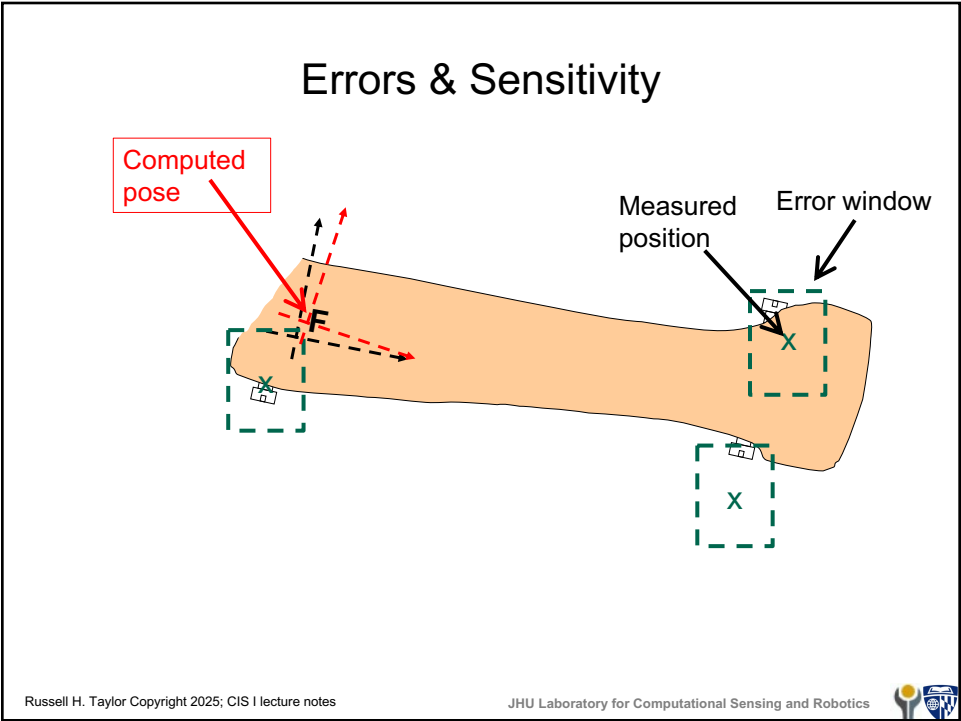
93



94



95



96

Errors & Sensitivity

Previous expression was

$$\Delta \vec{v}_1 \approx \mathbf{R} \bullet (\Delta \vec{b}_1 + \vec{\alpha} \times \vec{b} + \Delta \vec{p}_1)$$

Substituting triple product and rearranging gives

$$\Delta \vec{v}_1 \approx \left[\mathbf{R} \mid \mathbf{R} \mid \mathbf{R} \bullet \text{skew}(-\vec{b}) \right] \bullet \begin{bmatrix} \Delta \vec{b}_1 \\ \Delta \vec{p} \\ \vec{\alpha} \end{bmatrix}$$

So

$$\begin{bmatrix} -\varepsilon \\ -\varepsilon \\ -\varepsilon \end{bmatrix} \leq \left[\mathbf{R} \mid \mathbf{R} \mid \mathbf{R} \bullet \text{skew}(-\vec{b}) \right] \begin{bmatrix} \Delta \vec{b}_1 \\ \Delta \vec{p} \\ \vec{\alpha} \end{bmatrix} \leq \begin{bmatrix} \varepsilon \\ \varepsilon \\ \varepsilon \end{bmatrix}$$

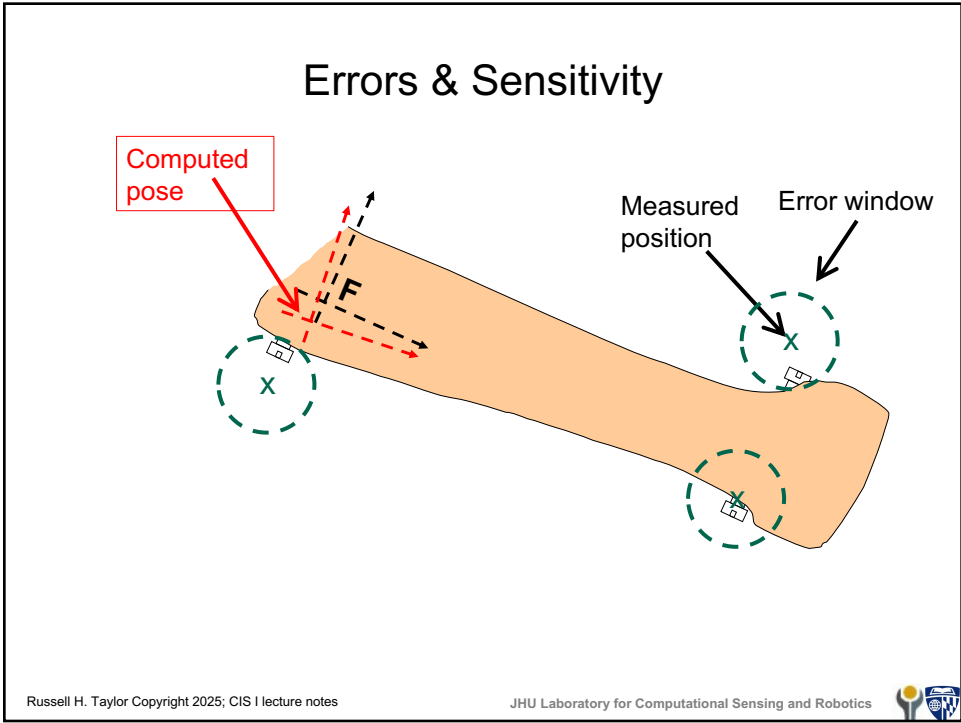


Errors & Sensitivity

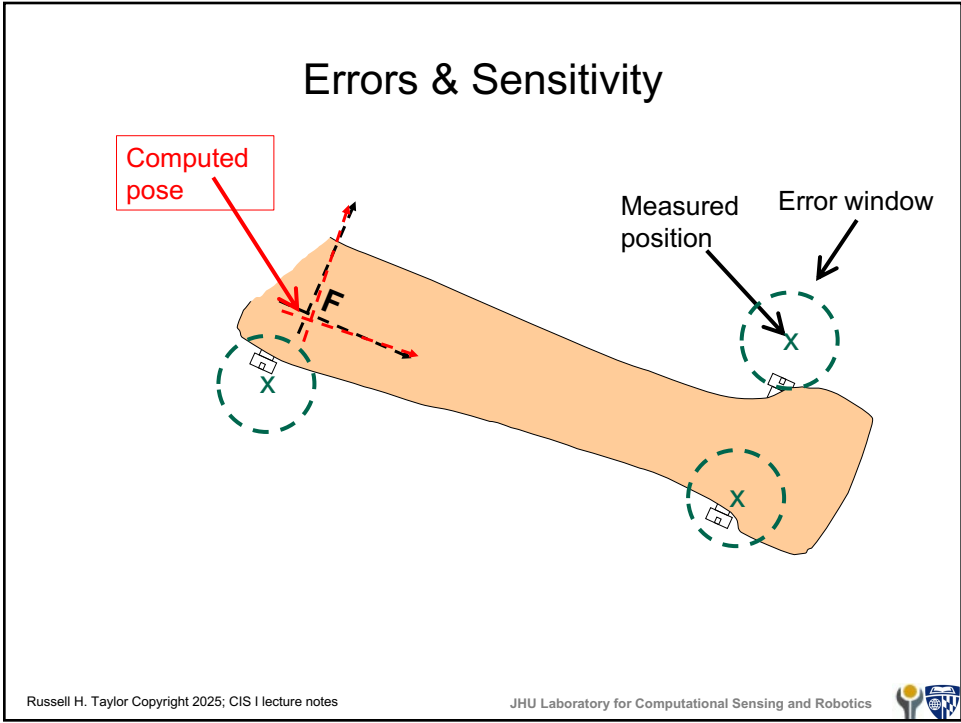
Now, suppose we know that $|\Delta \vec{b}_1| \leq \beta$, this will give us a system of linear constraints

$$\begin{bmatrix} -\varepsilon \\ -\varepsilon \\ -\varepsilon \\ -\beta \\ -\beta \\ -\beta \end{bmatrix} \leq \left[\begin{array}{c|c|c} \mathbf{R} & \mathbf{R} & \mathbf{R} \bullet \text{skew}(-\vec{b}) \\ \hline \mathbf{I} & \mathbf{0} & \mathbf{0} \end{array} \right] \begin{bmatrix} \Delta \vec{b}_1 \\ \Delta \vec{p}_1 \\ \vec{\alpha} \end{bmatrix} \leq \begin{bmatrix} \varepsilon \\ \varepsilon \\ \varepsilon \\ \beta \\ \beta \\ \beta \end{bmatrix}$$



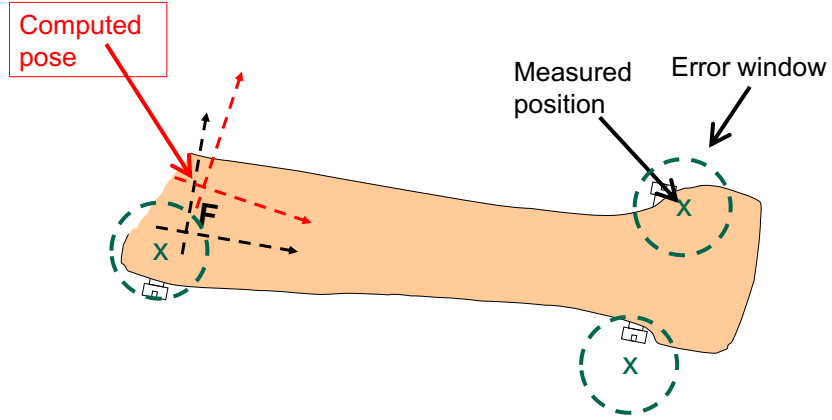


99



100

Errors & Sensitivity



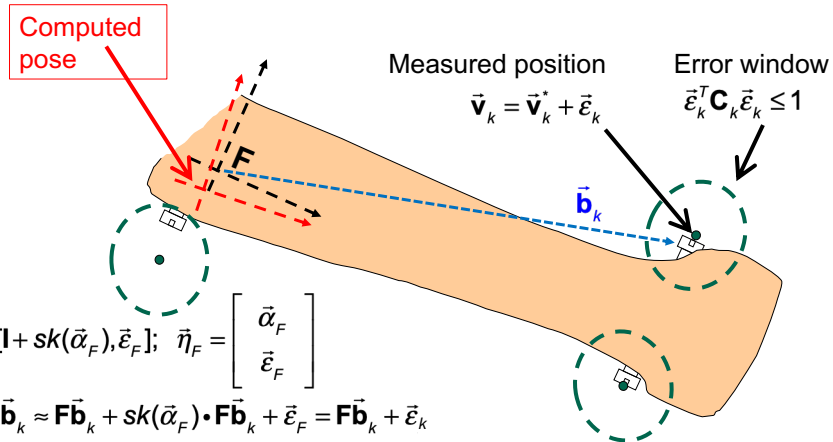
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101

Errors & Sensitivity



$$\Delta \mathbf{F} \approx [\mathbf{I} + sk(\vec{\alpha}_F), \vec{e}_F]; \quad \vec{\eta}_F = \begin{bmatrix} \vec{\alpha}_F \\ \vec{e}_F \end{bmatrix}$$

$$\Delta \mathbf{F} \cdot \mathbf{F} \vec{b}_k \approx \mathbf{F} \vec{b}_k + sk(\vec{\alpha}_F) \cdot \mathbf{F} \vec{b}_k + \vec{e}_F = \mathbf{F} \vec{b}_k + \vec{e}_k$$

$$\vec{e}_k = sk(\vec{\alpha}_F) \cdot \mathbf{F} \vec{b}_k + \vec{e}_F = \begin{bmatrix} -sk(\mathbf{F} \vec{b}_k) & \mathbf{I} \end{bmatrix} \cdot \vec{\eta}_F = \mathbf{M}_k \vec{\eta}_F$$

$$\vec{\eta}_F^T \mathbf{M}_k^T \mathbf{C}_k \mathbf{M}_k \vec{\eta}_F \leq 1$$

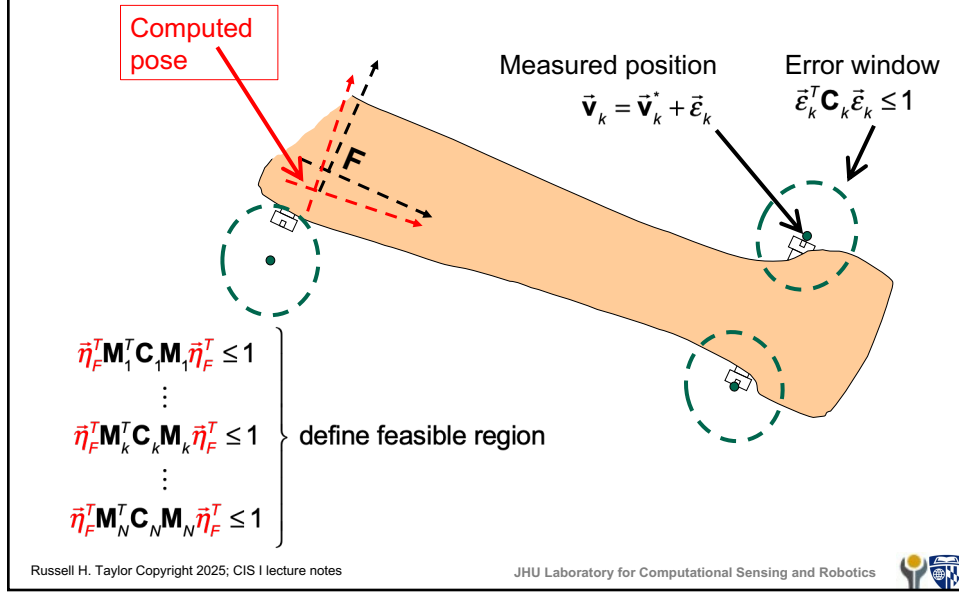
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102

Errors & Sensitivity



103

Error from frame composition

Consider $\mathbf{R}_1^* \mathbf{R}_2^* = \mathbf{R}_3^*$ where $\mathbf{R}_1^* = \mathbf{R}_1 \Delta \mathbf{R}_1$, $\mathbf{R}_2^* = \mathbf{R}_2 \Delta \mathbf{R}_2$, $\mathbf{R}_3^* = \mathbf{R}_3 \Delta \mathbf{R}_3$ and $\Delta \mathbf{R}_1 \approx \mathbf{I} + sk(\vec{\alpha}_1)$, $\Delta \mathbf{R}_2 \approx \mathbf{I} + sk(\vec{\alpha}_2)$, estimate $\Delta \mathbf{R}_3 \approx \mathbf{I} + sk(\vec{\alpha}_3)$

$$\mathbf{R}_1 \Delta \mathbf{R}_1 \mathbf{R}_2 \Delta \mathbf{R}_2 = \mathbf{R}_1 \mathbf{R}_2 \Delta \mathbf{R}_3$$

$$\mathbf{R}_1 (\mathbf{I} + sk(\vec{\alpha}_1)) \mathbf{R}_2 (\mathbf{I} + sk(\vec{\alpha}_2)) \approx \mathbf{R}_1 \mathbf{R}_2 (\mathbf{I} + sk(\vec{\alpha}_3))$$

$$(\mathbf{R}_1 \mathbf{R}_2)^{-1} \mathbf{R}_1 (\mathbf{I} + sk(\vec{\alpha}_1)) \mathbf{R}_2 (\mathbf{I} + sk(\vec{\alpha}_2)) \approx \mathbf{I} + sk(\vec{\alpha}_3)$$

$$\mathbf{R}_2^{-1} \mathbf{R}_1^{-1} \mathbf{R}_1 (\mathbf{I} + sk(\vec{\alpha}_1)) \mathbf{R}_2 (\mathbf{I} + sk(\vec{\alpha}_2)) \approx \mathbf{I} + sk(\vec{\alpha}_3)$$

$$\mathbf{I} + \mathbf{R}_2^{-1} sk(\vec{\alpha}_1) \mathbf{R}_2 + sk(\vec{\alpha}_2) + \mathbf{R}_2^{-1} sk(\vec{\alpha}_1) \mathbf{R}_2 sk(\vec{\alpha}_2) \approx \mathbf{I} + sk(\vec{\alpha}_3)$$

$$\mathbf{R}_2^{-1} sk(\vec{\alpha}_1) \mathbf{R}_2 + sk(\vec{\alpha}_2) \approx sk(\vec{\alpha}_3)$$

Since $\mathbf{R}^{-1} \cdot (\vec{a} \times \mathbf{R} \vec{b}) = (\mathbf{R}^{-1} \vec{a}) \times \vec{b}$ for all $\mathbf{R}, \vec{a}, \vec{b}$ we get $\mathbf{R}_2^{-1} sk(\vec{\alpha}_1) \mathbf{R}_2 = sk(\mathbf{R}_2^{-1} \vec{\alpha}_1)$

$$sk(\vec{\alpha}_3) \approx sk(\mathbf{R}_2^{-1} \vec{\alpha}_1) + sk(\vec{\alpha}_2) = sk(\mathbf{R}_2^{-1} \vec{\alpha}_1 + \vec{\alpha}_2)$$

$$\vec{\alpha}_3 \approx \mathbf{R}_2^{-1} \vec{\alpha}_1 + \vec{\alpha}_2$$

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104

Error from frame composition

Consider $\mathbf{F}_1^* \mathbf{F}_2^* = \mathbf{F}_3^*$ where $\mathbf{F}_1^* = \mathbf{F}_1 \Delta \mathbf{F}_1$, $\mathbf{F}_2^* = \mathbf{F}_2 \Delta \mathbf{F}_2$, $\mathbf{F}_3^* = \mathbf{F}_3 \Delta \mathbf{F}_3$
 and $\Delta \mathbf{F}_1 \approx [\mathbf{I} + sk(\vec{\alpha}_1), \vec{\epsilon}_1]$, $\Delta \mathbf{F}_2 \approx [\mathbf{I} + sk(\vec{\alpha}_2), \vec{\epsilon}_2]$,
 estimate $\Delta \mathbf{F}_3 \approx [\mathbf{I} + sk(\vec{\alpha}_3), \vec{\epsilon}_3]$

From before, we have $\vec{\alpha}_3 \approx \mathbf{R}_2^{-1} \vec{\alpha}_1 + \vec{\alpha}_2$. So now we just need $\vec{\epsilon}_3$.

$$\begin{aligned} \mathbf{F}_3 \Delta \mathbf{F}_3 &= [\mathbf{R}_3 \Delta \mathbf{R}_3, \mathbf{R}_3 \Delta \vec{p}_3 + \vec{p}_3] \\ \vec{p}_3 + \mathbf{R}_3 \Delta \vec{p}_3 &= \mathbf{R}_1 (\Delta \mathbf{R}_1 (\vec{p}_2 + \mathbf{R}_2 \vec{\epsilon}_2) + \vec{\epsilon}_1) + \vec{p}_1 \\ \vec{p}_3 + \mathbf{R}_3 \vec{\epsilon}_3 &\approx \mathbf{R}_1 (\mathbf{I} + sk(\vec{\alpha}_1)) (\vec{p}_2 + \mathbf{R}_2 \vec{\epsilon}_2) + \mathbf{R}_1 \vec{\epsilon}_1 + \vec{p}_1 \\ &= \mathbf{R}_1 \vec{p}_2 + \mathbf{R}_1 \mathbf{R}_2 \vec{\epsilon}_2 + \mathbf{R}_1 \cdot (\vec{\alpha}_1 \times \vec{p}_2 + \vec{\alpha}_1 \times \mathbf{R}_2 \vec{\epsilon}_2) + \vec{p}_1 + \mathbf{R}_1 \vec{\epsilon}_1 \\ &= \vec{p}_3 + \mathbf{R}_1 \mathbf{R}_2 \vec{\epsilon}_2 + \mathbf{R}_1 \cdot (\vec{\alpha}_1 \times \vec{p}_2 + \vec{\alpha}_1 \times \mathbf{R}_2 \vec{\epsilon}_2) + \mathbf{R}_1 \vec{\epsilon}_1 \\ \mathbf{R}_3 \vec{\epsilon}_3 &\approx \mathbf{R}_1 \mathbf{R}_2 \vec{\epsilon}_2 + \mathbf{R}_1 \cdot \vec{\alpha}_1 \times \vec{p}_2 + \mathbf{R}_1 \vec{\epsilon}_1 \\ \vec{\epsilon}_3 &\approx \mathbf{R}_2^{-1} \mathbf{R}_1^{-1} \mathbf{R}_1 \mathbf{R}_2 \vec{\epsilon}_2 + \mathbf{R}_2^{-1} \mathbf{R}_1^{-1} \cdot \vec{\alpha}_1 \times \vec{p}_2 + \mathbf{R}_2^{-1} \mathbf{R}_1^{-1} \mathbf{R}_1 \vec{\epsilon}_1 \\ &= \vec{\epsilon}_2 + \mathbf{R}_2^{-1} \cdot \vec{\alpha}_1 \times \vec{p}_2 + \mathbf{R}_2^{-1} \vec{\epsilon}_1 \\ &= \vec{\epsilon}_2 - \mathbf{R}_2^{-1} sk(\vec{p}_2) \vec{\alpha}_1 + \mathbf{R}_2^{-1} \vec{\epsilon}_1 \end{aligned}$$

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106

Inverse of frame transformation with errors

For $\mathbf{F}^* = \mathbf{F} \Delta \mathbf{F}$, if we want $\Delta \mathbf{F}_i$ such that $\mathbf{F}_i^* = \mathbf{F}_i \Delta \mathbf{F}_i$, we have

$$\begin{aligned} \mathbf{F}_i &= \mathbf{F}^{-1} = [\mathbf{R}^{-1}, -\mathbf{R}^{-1} \vec{p}] \\ \mathbf{F}_i^* &= \mathbf{F}_i \Delta \mathbf{F}_i = (\mathbf{F} \Delta \mathbf{F})^{-1} = \Delta \mathbf{F}^{-1} \mathbf{F}^{-1} = [\Delta \mathbf{R}^{-1}, -\Delta \mathbf{R}^{-1} \Delta \vec{p}] \cdot [\mathbf{R}^{-1}, -\mathbf{R}^{-1} \vec{p}] \\ \mathbf{F}_i \Delta \mathbf{F}_i &= [\Delta \mathbf{R}^{-1} \mathbf{R}^{-1}, -\Delta \mathbf{R}^{-1} \mathbf{R}^{-1} \vec{p} - \Delta \mathbf{R}^{-1} \Delta \vec{p}] \\ \Delta \mathbf{F}_i &= \mathbf{F}_i^{-1} [\Delta \mathbf{R}^{-1} \mathbf{R}^{-1}, -\Delta \mathbf{R}^{-1} \mathbf{R}^{-1} \vec{p} - \Delta \mathbf{R}^{-1} \Delta \vec{p}] \\ &= (\mathbf{F}^{-1})^{-1} [\Delta \mathbf{R}^{-1} \mathbf{R}^{-1}, -\Delta \mathbf{R}^{-1} \mathbf{R}^{-1} \vec{p} - \Delta \mathbf{R}^{-1} \Delta \vec{p}] \\ &= [\mathbf{R}, \vec{p}] \cdot [\Delta \mathbf{R}^{-1} \mathbf{R}^{-1}, -\Delta \mathbf{R}^{-1} \mathbf{R}^{-1} \vec{p} - \Delta \mathbf{R}^{-1} \Delta \vec{p}] \\ &= [\mathbf{R} \Delta \mathbf{R}^{-1} \mathbf{R}^{-1}, -\mathbf{R} \Delta \mathbf{R}^{-1} \mathbf{R}^{-1} \vec{p} - \mathbf{R} \Delta \mathbf{R}^{-1} \Delta \vec{p} + \vec{p}] \end{aligned}$$

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107

Inverse of frame transformation with errors

For $\mathbf{F}^* = \Delta \mathbf{F} \mathbf{F}$, if we want $\mathbf{F}_i^* = \Delta \mathbf{F} \mathbf{F}_i$, we have

$$\mathbf{F}_i = \mathbf{F}^{-1} = [\mathbf{R}^{-1}, -\mathbf{R}^{-1}\vec{\mathbf{p}}]$$

$$\mathbf{F}_i^* = \Delta \mathbf{F} \mathbf{F}_i = (\Delta \mathbf{F} \mathbf{F})^{-1}$$

$$= \mathbf{F}^{-1} \Delta \mathbf{F}^{-1}$$

$$= [\mathbf{R}^{-1}, -\mathbf{R}^{-1}\vec{\mathbf{p}}] \cdot [\Delta \mathbf{R}^{-1}, -\Delta \mathbf{R}^{-1} \Delta \vec{\mathbf{p}}]$$

$$\Delta \mathbf{F} \mathbf{F}_i = [\mathbf{R}^{-1} \Delta \mathbf{R}^{-1}, -\mathbf{R}^{-1} \Delta \mathbf{R}^{-1} \Delta \vec{\mathbf{p}} - \mathbf{R}^{-1} \vec{\mathbf{p}}]$$

$$\Delta \mathbf{F}_i = [\mathbf{R}^{-1} \Delta \mathbf{R}^{-1}, -\mathbf{R}^{-1} \Delta \mathbf{R}^{-1} \Delta \vec{\mathbf{p}} - \mathbf{R}^{-1} \vec{\mathbf{p}}] \cdot (\mathbf{F}^{-1})^{-1}$$

$$= [\mathbf{R}^{-1} \Delta \mathbf{R}^{-1}, -\mathbf{R}^{-1} \Delta \mathbf{R}^{-1} \Delta \vec{\mathbf{p}} - \mathbf{R}^{-1} \vec{\mathbf{p}}] \cdot [\mathbf{R}, \vec{\mathbf{p}}]$$

$$= [\mathbf{R}^{-1} \Delta \mathbf{R}^{-1} \mathbf{R}, \mathbf{R}^{-1} \Delta \mathbf{R}^{-1} \vec{\mathbf{p}} - \mathbf{R}^{-1} \Delta \mathbf{R}^{-1} \Delta \vec{\mathbf{p}} - \mathbf{R}^{-1} \vec{\mathbf{p}}]$$



Inverse of frame transformation with errors

Suppose we know that $\Delta \mathbf{R}$ is "small", i.e., $\Delta \mathbf{R} \approx \mathbf{I} + sk(\vec{\alpha})$, and for notational convenience we write $\Delta \vec{\mathbf{p}} = \vec{\epsilon}$, we get

$$\Delta \mathbf{R}_i = \mathbf{R} \Delta \mathbf{R}^{-1} \mathbf{R}^{-1} \approx \mathbf{R} (\mathbf{I} + sk(\vec{\alpha}))^{-1} \mathbf{R}^{-1}$$

$$\approx \mathbf{R} (\mathbf{I} - sk(\vec{\alpha})) \mathbf{R}^{-1}$$

$$= \mathbf{R} \mathbf{R}^{-1} - \mathbf{R} sk(\vec{\alpha}) \mathbf{R}^{-1}$$

$$= \mathbf{I} - \mathbf{R} sk(\vec{\alpha}) \mathbf{R}^{-1}$$

$$= \mathbf{I} - sk \left((\mathbf{R}^{-1})^{-1} \vec{\alpha} \right) = \mathbf{I} - sk(\mathbf{R} \vec{\alpha}) = \mathbf{I} + sk(-\mathbf{R} \vec{\alpha})$$

$$\Delta \vec{\mathbf{p}}_i = -\mathbf{R} \Delta \mathbf{R}^{-1} \mathbf{R}^{-1} \vec{\mathbf{p}} - \mathbf{R} \Delta \mathbf{R}^{-1} \Delta \vec{\mathbf{p}} + \vec{\mathbf{p}}$$

$$\approx -\mathbf{R} (\mathbf{I} - sk(\vec{\alpha})) \mathbf{R}^{-1} \vec{\mathbf{p}} - \mathbf{R} (\mathbf{I} - sk(\vec{\alpha})) \vec{\epsilon} + \vec{\mathbf{p}}$$

$$= -\vec{\mathbf{p}} + \mathbf{R} sk(\vec{\alpha}) \mathbf{R}^{-1} \vec{\mathbf{p}} - \mathbf{R} \vec{\epsilon} + \mathbf{R} sk(\vec{\alpha}) \vec{\epsilon} + \vec{\mathbf{p}}$$

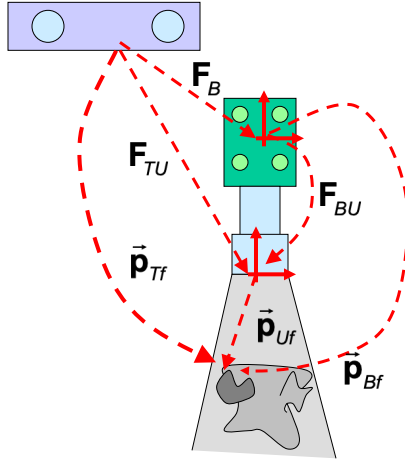
$$\approx \mathbf{R} sk(\vec{\alpha}) \mathbf{R}^{-1} \vec{\mathbf{p}} - \mathbf{R} \vec{\epsilon} = \mathbf{R} (\vec{\alpha} \times \mathbf{R}^{-1} \vec{\mathbf{p}}) - \mathbf{R} \vec{\epsilon}$$

$$\Delta \vec{\mathbf{p}} \approx -\mathbf{R} \vec{\epsilon} - \mathbf{R} sk(\mathbf{R}^{-1} \vec{\mathbf{p}}) \vec{\alpha}$$

$$\vec{\alpha}_i = -\mathbf{R} \vec{\alpha} \quad \vec{\epsilon}_i = -\mathbf{R} \vec{\epsilon} - \mathbf{R} sk(\mathbf{R}^{-1} \vec{\mathbf{p}}) \vec{\alpha}$$



Example



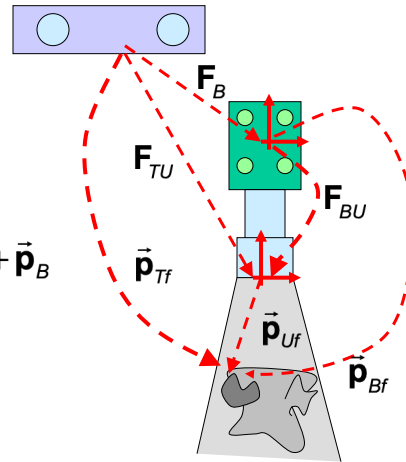
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111

Example



$$\vec{p}_{Tf} = \mathbf{F}_{TU} \cdot \vec{p}_{Uf}$$

$$\begin{aligned} \mathbf{F}_{TU} &= \mathbf{F}_B \cdot \mathbf{F}_{BU} \\ &= [\mathbf{R}_B \cdot \mathbf{R}_{BU}, \mathbf{R}_B \cdot \vec{p}_{BU} + \vec{p}_B] \end{aligned}$$

$$\vec{p}_{Tf} = \mathbf{R}_B \cdot \mathbf{R}_{BU} \cdot \vec{p}_{Uf} + \mathbf{R}_B \cdot \vec{p}_{BU} + \vec{p}_B$$

Also

$$\vec{p}_{Tf} = \mathbf{F}_B \cdot \vec{p}_{Bf}$$

$$\begin{aligned} \vec{p}_{Bf} &= \mathbf{F}_{BU} \cdot \vec{p}_{Uf} \\ &= \mathbf{R}_{BU} \cdot \vec{p}_{Uf} + \vec{p}_{BU} \end{aligned}$$

$$\vec{p}_{Tf} = \mathbf{R}_B \cdot \mathbf{R}_{BU} \cdot \vec{p}_{Uf} + \mathbf{R}_B \cdot \vec{p}_{BU} + \vec{p}_B$$

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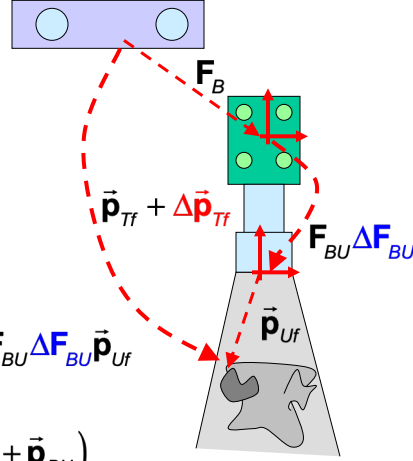
112

Example

Suppose that the track body to US calibration is not perfect

$$\begin{aligned}\mathbf{F}_{BU}^* &= \mathbf{F}_{BU} \Delta \mathbf{F}_{BU} \\ &= [\mathbf{R}_{BU} \Delta \mathbf{R}_{BU}, \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU} + \vec{\mathbf{p}}_{BU}]\end{aligned}$$

$$\begin{aligned}\vec{\mathbf{p}}_{Bf}^* &= \mathbf{F}_{BU}^* \bullet \vec{\mathbf{p}}_{Uf} \quad \text{i.e.,} \quad \vec{\mathbf{p}}_{Bf} + \Delta \vec{\mathbf{p}}_{Bf} = \mathbf{F}_{BU} \Delta \mathbf{F}_{BU} \vec{\mathbf{p}}_{Uf} \\ \Delta \vec{\mathbf{p}}_{Bf} &= \mathbf{F}_{BU} \Delta \mathbf{F}_{BU} \vec{\mathbf{p}}_{Uf} - \vec{\mathbf{p}}_{Bf} \\ &= \mathbf{F}_{BU} (\Delta \mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf} + \Delta \vec{\mathbf{p}}_{BU}) - (\mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf} + \vec{\mathbf{p}}_{BU}) \\ &= \mathbf{R}_{BU} \Delta \mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf} + \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU} + \vec{\mathbf{p}}_{BU} - \mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf} - \vec{\mathbf{p}}_{BU} \\ &= \mathbf{R}_{BU} \Delta \mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf} + \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU} - \mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf}\end{aligned}$$



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113

Example

Continuing ...

$$\begin{aligned}\Delta \vec{\mathbf{p}}_{Bf} &= \mathbf{R}_{BU} \Delta \mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf} + \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU} - \mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf} \\ &\approx \mathbf{R}_{BU} (\mathbf{I} + \text{skew}(\vec{\alpha}_{BU})) \vec{\mathbf{p}}_{Uf} + \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU} - \mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf} \\ &= \cancel{\mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf}} + \mathbf{R}_{BU} \bullet \vec{\alpha}_{BU} \times \vec{\mathbf{p}}_{Uf} + \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU} - \cancel{\mathbf{R}_{BU} \vec{\mathbf{p}}_{Uf}} \\ &= \mathbf{R}_{BU} \bullet \vec{\alpha}_{BU} \times \vec{\mathbf{p}}_{Uf} + \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU} \\ &= -\mathbf{R}_{BU} \bullet \vec{\mathbf{p}}_{Uf} \times \vec{\alpha}_{BU} + \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU} \\ &= \mathbf{R}_{BU} \text{skew}(-\vec{\mathbf{p}}_{Uf}) \vec{\alpha}_{BU} + \mathbf{R}_{BU} \Delta \vec{\mathbf{p}}_{BU}\end{aligned}$$

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114

Example

$$\begin{aligned}
 \vec{p}_{Tf} + \Delta \vec{p}_{Tf} &= \mathbf{F}_B \Delta \mathbf{F}_B (\vec{p}_{Bf} + \Delta \vec{p}_{Bf}) \\
 \Delta \vec{p}_{Tf} &= \mathbf{F}_B \Delta \mathbf{F}_B (\vec{p}_{Bf} + \Delta \vec{p}_{Bf}) - \mathbf{F}_B \vec{p}_{Bf} \\
 \Delta \mathbf{F}_B (\vec{p}_{Bf} + \Delta \vec{p}_{Bf}) &= \Delta \mathbf{R}_B (\vec{p}_{Bf} + \Delta \vec{p}_{Bf}) + \Delta \vec{p}_B \\
 &\approx (\mathbf{I} + \text{skew}(\vec{\alpha}_B)) (\vec{p}_{Bf} + \Delta \vec{p}_{Bf}) + \Delta \vec{p}_B \\
 &= (\vec{p}_{Bf} + \Delta \vec{p}_{Bf}) + \vec{\alpha}_B \times \vec{p}_{Bf} + \cancel{\vec{\alpha}_B \times \Delta \vec{p}_{Bf}} + \Delta \vec{p}_B \\
 &\approx \vec{p}_{Bf} + \Delta \vec{p}_{Bf} + \vec{\alpha}_B \times \vec{p}_{Bf} + \Delta \vec{p}_B \\
 \Delta \vec{p}_{Tf} &\approx \mathbf{F}_B (\vec{p}_{Bf} + \Delta \vec{p}_{Bf} + \vec{\alpha}_B \times \vec{p}_{Bf} + \Delta \vec{p}_B) - \mathbf{F}_B \vec{p}_{Bf} \\
 &= \mathbf{R}_B (\vec{p}_{Bf} + \Delta \vec{p}_{Bf} + \vec{\alpha}_B \times \vec{p}_{Bf} + \Delta \vec{p}_B) + \vec{p}_B - (\mathbf{R}_B \vec{p}_{Bf} + \vec{p}_B) \\
 &= \mathbf{R}_B (\Delta \vec{p}_{Bf} + \vec{\alpha}_B \times \vec{p}_{Bf} + \Delta \vec{p}_B) \\
 \Delta \vec{p}_{Bf} &\approx \mathbf{R}_{BU} \text{skew}(-\vec{p}_{BU}) \vec{\alpha}_{BU} + \mathbf{R}_{BU} \Delta \vec{p}_{BU}
 \end{aligned}$$

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115

Example

$$\begin{aligned}
 \Delta \vec{p}_{Tf} &\approx \mathbf{R}_B (\Delta \vec{p}_{Bf} + \vec{\alpha}_B \times \vec{p}_{Bf} + \Delta \vec{p}_B) \\
 \Delta \vec{p}_{Bf} &\approx \mathbf{R}_{BU} \text{skew}(-\vec{p}_{BU}) \vec{\alpha}_{BU} + \mathbf{R}_{BU} \Delta \vec{p}_{BU} \\
 \Delta \vec{p}_{Tf} &\approx \mathbf{R}_B (\mathbf{R}_{BU} \text{skew}(-\vec{p}_{BU}) \vec{\alpha}_{BU} + \mathbf{R}_{BU} \Delta \vec{p}_{BU} + \vec{\alpha}_B \times \vec{p}_{Bf} + \Delta \vec{p}_B) \\
 &= \begin{pmatrix} \mathbf{R}_B \mathbf{R}_{BU} \text{skew}(-\vec{p}_{BU}) \vec{\alpha}_{BU} + \mathbf{R}_B \mathbf{R}_{BU} \Delta \vec{p}_{BU} \\ + \mathbf{R}_B \text{skew}(-\vec{p}_{Bf}) \vec{\alpha}_B + \mathbf{R}_B \Delta \vec{p}_B \end{pmatrix} \\
 &= \begin{bmatrix} \mathbf{R}_B \mathbf{R}_{BU} \text{skew}(-\vec{p}_{BU}) & \mathbf{R}_B \mathbf{R}_{BU} & \mathbf{R}_B \text{skew}(-\vec{p}_{Bf}) & \mathbf{R}_B \end{bmatrix} \begin{bmatrix} \vec{\alpha}_{BU} \\ \Delta \vec{p}_{BU} \\ \vec{\alpha}_B \\ \Delta \vec{p}_B \end{bmatrix}
 \end{aligned}$$

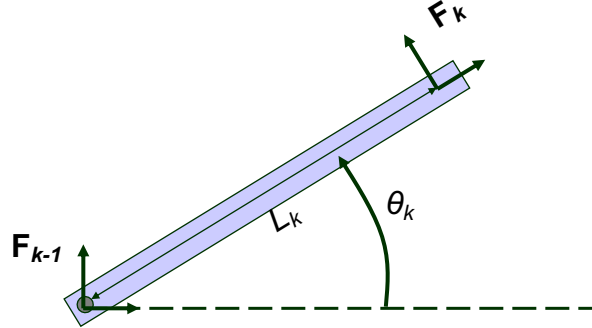
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116

Error Propagation in Kinematic Chains



$$\begin{aligned}
 \mathbf{F}_k^* &= \mathbf{F}_{k-1}^* \bullet \mathbf{F}_{k-1,k}^* \\
 \mathbf{F}_k \Delta \mathbf{F}_k &= \mathbf{F}_{k-1} \Delta \mathbf{F}_{k-1} \mathbf{F}_{k-1,k} \Delta \mathbf{F}_{k-1,k} \\
 \Delta \mathbf{F}_k &= (\mathbf{F}_k^{-1} \mathbf{F}_{k-1}) \Delta \mathbf{F}_{k-1} \mathbf{F}_{k-1,k} \Delta \mathbf{F}_{k-1,k} \\
 &= (\mathbf{F}_{k-1,k}^{-1} \Delta \mathbf{F}_{k-1} \mathbf{F}_{k-1,k}) \Delta \mathbf{F}_{k-1,k}
 \end{aligned}$$

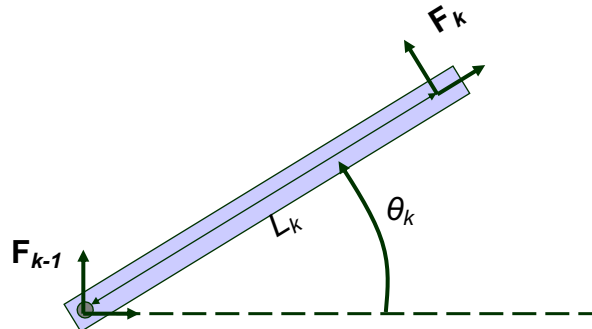
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117

Error Propagation in Kinematic Chains



$$\begin{aligned}
 \Delta \mathbf{F}_k &= (\mathbf{F}_k^{-1} \mathbf{F}_{k-1}) \Delta \mathbf{F}_{k-1} \mathbf{F}_{k-1,k} \Delta \mathbf{F}_{k-1,k} \\
 &= (\mathbf{F}_{k-1,k}^{-1} \Delta \mathbf{F}_{k-1} \mathbf{F}_{k-1,k}) \Delta \mathbf{F}_{k-1,k} \\
 \Delta \mathbf{R}_k &= (\mathbf{R}_{k-1,k}^{-1} \Delta \mathbf{R}_{k-1} \mathbf{R}_{k-1,k}) \Delta \mathbf{R}_{k-1,k} \\
 &\approx (\mathbf{R}_{k-1,k}^{-1} (\mathbf{I} + \text{skew}(\vec{\alpha}_{k-1})) \mathbf{R}_{k-1,k}) (\mathbf{I} + \text{skew}(\vec{\alpha}_{k-1,k})) \\
 &\approx \mathbf{I} + (\mathbf{R}_{k-1,k}^{-1} \text{skew}(\vec{\alpha}_{k-1}) \mathbf{R}_{k-1,k}) + \text{skew}(\vec{\alpha}_{k-1,k}) = \mathbf{I} + \text{skew}(\mathbf{R}_{k-1,k}^{-1} \vec{\alpha}_{k-1} + \vec{\alpha}_{k-1,k})
 \end{aligned}$$

Note: This is same as what we could have obtained by substituting in formulas from the "error from frame composition" slides given earlier.

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118

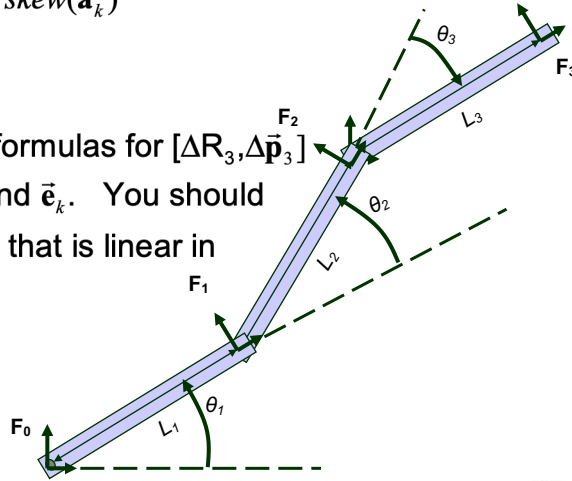
Exercise

Suppose that you have

$$\Delta \mathbf{R}_{k-1,k} = \Delta \mathbf{R}(\vec{\mathbf{a}}_k) \cong \mathbf{I} + \text{skew}(\vec{\mathbf{a}}_k)$$

$$\Delta \vec{\mathbf{p}}_{k-1,k} = \vec{\mathbf{e}}_k$$

Work out approximate formulas for $[\Delta \mathbf{R}_3, \Delta \vec{\mathbf{p}}_3]$ in terms of $L_k, \vec{\mathbf{r}}_k, \theta_k, \vec{\mathbf{a}}_k$ and $\vec{\mathbf{e}}_k$. You should come up with a formula that is linear in $L_k, \vec{\mathbf{a}}_k$, and $\vec{\mathbf{e}}_k$.



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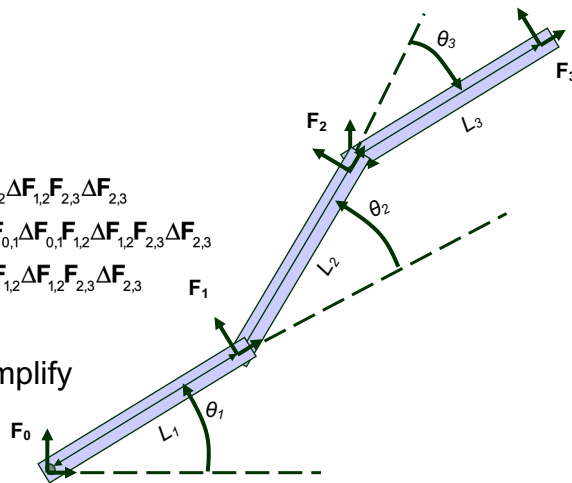
119

Exercise

Suppose we want to know error in $\mathbf{F}_{0,3} = \mathbf{F}_0^{-1} \mathbf{F}_3$

$$\begin{aligned} \mathbf{F}_{0,3} &= \mathbf{F}_0^{-1} \mathbf{F}_{0,1} \mathbf{F}_{1,2} \mathbf{F}_{2,3} \\ \mathbf{F}_{0,3}^* &= \mathbf{F}_{0,1}^* \mathbf{F}_{1,2}^* \mathbf{F}_{2,3}^* \\ \mathbf{F}_{0,3} \Delta \mathbf{F}_{0,3} &= \mathbf{F}_{0,1}^* \mathbf{F}_{1,2}^* \mathbf{F}_{2,3}^* \\ \Delta \mathbf{F}_{0,3} &= \mathbf{F}_{0,3}^{-1} \mathbf{F}_{0,1} \Delta \mathbf{F}_{0,1} \mathbf{F}_{1,2} \mathbf{F}_{2,3} \Delta \mathbf{F}_{2,3} \\ &= \mathbf{F}_{2,3}^{-1} \mathbf{F}_{1,2}^{-1} \mathbf{F}_{0,1}^{-1} \mathbf{F}_{0,1} \Delta \mathbf{F}_{0,1} \mathbf{F}_{1,2} \mathbf{F}_{2,3} \Delta \mathbf{F}_{2,3} \\ &= \mathbf{F}_{2,3}^{-1} \mathbf{F}_{1,2}^{-1} \Delta \mathbf{F}_{0,1} \mathbf{F}_{1,2} \mathbf{F}_{2,3} \Delta \mathbf{F}_{2,3} \end{aligned}$$

Now substitute and simplify



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120

Parametric Sensitivity

Suppose you have an explicit formula like

$$\vec{p}_3 = \begin{bmatrix} L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3) \\ 0 \end{bmatrix}$$

and know that the only variation is in parameters like L_k and θ_k . Then you can estimate the variation in $\vec{p}_3$ as a function of variation in L_k and θ_k by remembering your calculus.

$$\Delta \vec{p}_3 \cong \begin{bmatrix} \frac{\partial \vec{p}_3}{\partial \vec{L}} & \frac{\partial \vec{p}_3}{\partial \vec{\theta}} \end{bmatrix} \begin{bmatrix} \Delta \vec{L} \\ \Delta \vec{\theta} \end{bmatrix}$$



Parametric Sensitivity

Grinding this out gives:

$$\Delta \vec{p}_3 \cong \begin{bmatrix} \frac{\partial \vec{p}_3}{\partial \vec{L}} & \frac{\partial \vec{p}_3}{\partial \vec{\theta}} \end{bmatrix} \begin{bmatrix} \Delta \vec{L} \\ \Delta \vec{\theta} \end{bmatrix}$$

where

$$\vec{L} = [L_1, L_2, L_3]^T$$

$$\vec{\theta} = [\theta_1, \theta_2, \theta_3]^T$$

$$\frac{\partial \vec{p}_3}{\partial \vec{L}} = \begin{bmatrix} \cos(\theta_1) & \cos(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2 + \theta_3) \\ \sin(\theta_1) & \sin(\theta_1 + \theta_2) & \sin(\theta_1 + \theta_2 + \theta_3) \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial \vec{p}_3}{\partial \vec{\theta}} = \begin{bmatrix} -L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2) - L_3 \sin(\theta_1 + \theta_2 + \theta_3) & -L_2 \sin(\theta_1 + \theta_2) - L_3 \sin(\theta_1 + \theta_2 + \theta_3) & -L_3 \sin(\theta_1 + \theta_2 + \theta_3) \\ L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3) & L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3) & L_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ 0 & 0 & 0 \end{bmatrix}$$



More generally ...

Suppose that we have a vector function

$$\vec{v} = \vec{g}(\vec{q}) = [g_1(\vec{q}), \dots, g_m(\vec{q})]^T$$

of parameters $\vec{q} = [q_1, \dots, q_n]$. Then we can estimate the value of

$$\vec{v} + \Delta \vec{v} \approx \vec{g}(\vec{q} + \Delta \vec{q})$$

by

$$\vec{v} + \Delta \vec{v} \approx \vec{g}(\vec{q}) + \mathbf{J}_g(\vec{q}) \bullet \Delta \vec{q}$$

where

$$\mathbf{J}_g(\vec{q}) = \begin{bmatrix} \frac{\partial g_1}{\partial q_1} & \frac{\partial g_1}{\partial q_2} & \dots & \frac{\partial g_1}{\partial q_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_i}{\partial q_1} & \dots & \frac{\partial g_i}{\partial q_j} & \dots & \frac{\partial g_i}{\partial q_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial q_1} & \frac{\partial g_m}{\partial q_2} & \dots & \frac{\partial g_m}{\partial q_n} \end{bmatrix}$$

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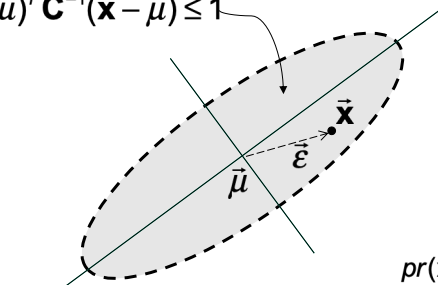
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123

Probabilistic Error Modeling: Multivariable Gaussian

$$(\vec{x} - \vec{\mu})^T \mathbf{C}^{-1} (\vec{x} - \vec{\mu}) \leq 1$$



If $\vec{x} \sim N(\vec{\mu}_x, \mathbf{C}_{xx})$, then

$$\vec{\epsilon} = \vec{x} - \vec{\mu} \sim N(\vec{0}, \mathbf{C})$$

$$\mathbf{A}\vec{x} + \vec{c} \sim N(\mathbf{A}\vec{\mu}_x + \vec{c}, \mathbf{A}\mathbf{C}_{xx}\mathbf{A}^T) \text{ for constants } \mathbf{A}, \vec{c}$$

Also there will be a random vector $\vec{\theta}$ with independent elements

$$\vec{\theta}_i \sim N(0, 1) \text{ and matrix } \mathbf{M} \text{ such that } \vec{x} = \vec{\mu} + \mathbf{M}\vec{\theta}, \text{ and}$$

$$\mathbf{C}_{xx} = \mathbf{M}\mathbf{M}^T = \mathbf{Q}\mathbf{\Lambda}^2\mathbf{Q}^T \text{ where } \mathbf{Q}\mathbf{Q}^T = \mathbf{I}, \mathbf{\Lambda} = \text{diag}(\vec{\lambda}), \mathbf{M} = \mathbf{Q}\text{diag}(\vec{\lambda})$$

$$\vec{x} \sim N(\vec{\mu}, \mathbf{C})$$

$$\vec{\mu} = E[\vec{x}]$$

$$\begin{aligned} \mathbf{C} &= E[(\vec{x} - \vec{\mu})(\vec{x} - \vec{\mu})^T] \\ &= E[\vec{x}\vec{x}^T] - \vec{\mu}\vec{\mu}^T \end{aligned}$$

$$pr(\vec{x}) = \frac{\exp(-(\vec{x} - \vec{\mu})^T \mathbf{C}^{-1} (\vec{x} - \vec{\mu}) / 2)}{\sqrt{(2\pi)^n |\mathbf{C}|}}$$

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124

Mahalanobis Distance

Given an n -dimensional random variable $\bar{\mathbf{x}}$ with mean $\bar{\mu}$ and positive-definite covariance $\mathbf{C}$, the Mahalanobis distance

$$d_M(\bar{\mathbf{x}}; \bar{\mu}, \mathbf{C}) = \sqrt{(\bar{\mathbf{x}} - \bar{\mu})^T \mathbf{C}^{-1} (\bar{\mathbf{x}} - \bar{\mu})}$$

is a random variable. Further $d_M(\bar{\mathbf{x}}; \bar{\mu}, \mathbf{C})^2$ follows a chi-squared distribution with n degrees of freedom. I.e.

$$d_M(\bar{\mathbf{x}}; \bar{\mu}, \mathbf{C})^2 \sim \chi^2(n)$$

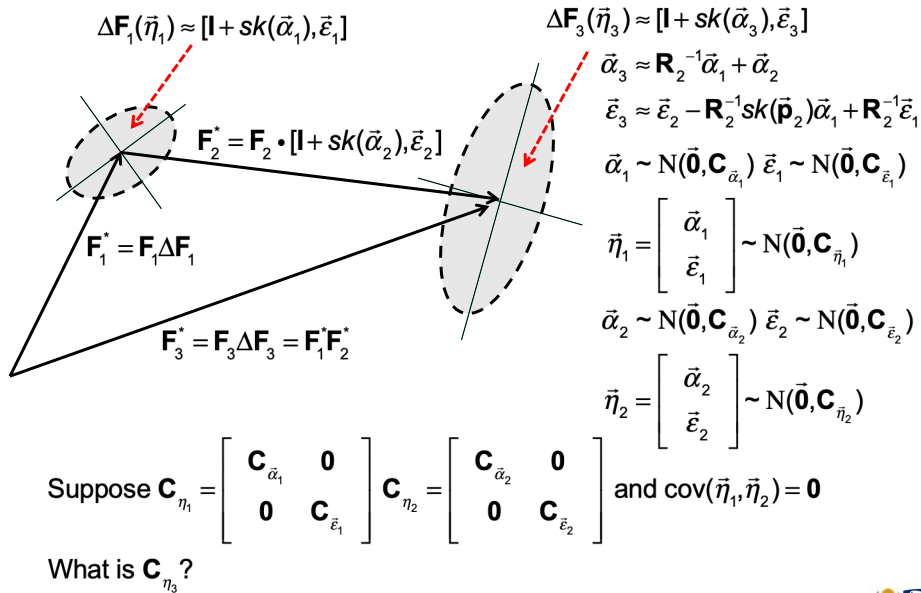
Further, the probability that $d_M^2 \leq \rho^2$ is given by the cumulative distribution function

$$\Pr[d_M^2 \leq \rho^2] = \frac{1}{\Gamma(n/2)} \gamma\left(\frac{n}{2}, \frac{\rho^2}{2}\right) \text{ where } \gamma(s, x) = \int_0^x t^{s-1} e^{-t} dt$$

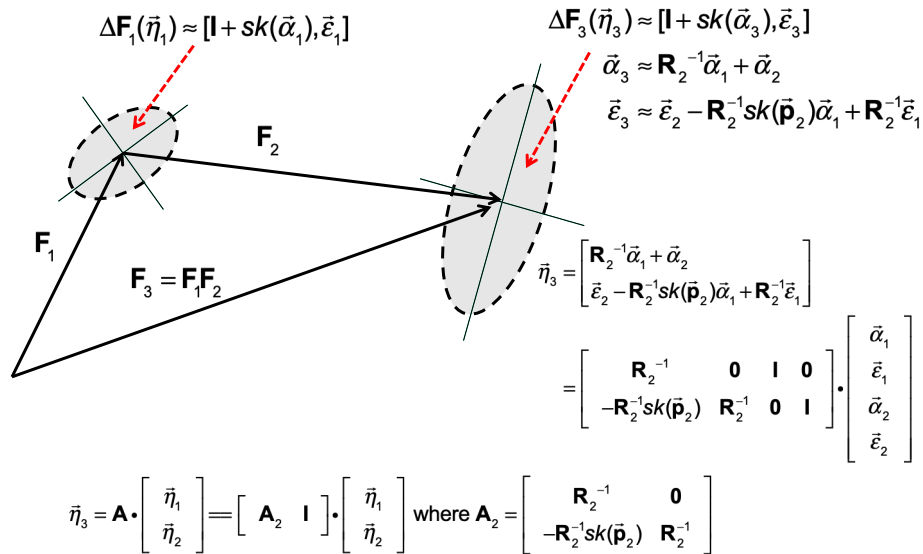
Note that math libraries typically have functions like "chi2cdf(s,n)", so you don't need to get too deep into the weeds to use this.



Transformation Composition Example



Transformation Composition Example



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127

Transformation Composition Example

$$\begin{aligned}\vec{\eta}_3 &= \mathbf{A} \cdot \begin{bmatrix} \vec{\eta}_1 \\ \vec{\eta}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_2 & \mathbf{I} \end{bmatrix} \cdot \begin{bmatrix} \vec{\eta}_1 \\ \vec{\eta}_2 \end{bmatrix} \text{ where } \mathbf{A}_2 = \begin{bmatrix} \mathbf{R}_2^{-1} & \mathbf{0} \\ -\mathbf{R}_2^{-1} sk(\vec{\mathbf{p}}_2) & \mathbf{R}_2^{-1} \end{bmatrix} \\ \text{cov}(\vec{\eta}_3) &= \begin{bmatrix} \mathbf{A}_2 & \mathbf{I} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{C}_{\vec{\eta}_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{\vec{\eta}_2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{A}_2^T \\ \mathbf{I} \end{bmatrix} = \mathbf{A}_2 \mathbf{C}_{\vec{\eta}_1} \mathbf{A}_2^T + \mathbf{C}_{\vec{\eta}_2} \\ \mathbf{A}_2 \mathbf{C}_{\vec{\eta}_1} \mathbf{A}_2^T &= \begin{bmatrix} \mathbf{R}_2^T & \mathbf{0} \\ \mathbf{R}_2^T sk(-\vec{\mathbf{p}}_2) & \mathbf{R}_2^T \end{bmatrix} \cdot \begin{bmatrix} \mathbf{C}_{\vec{\alpha}_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{\vec{\epsilon}_1} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{R}_2 & sk(\vec{\mathbf{p}}_2) \mathbf{R}_2 \\ \mathbf{0} & \mathbf{R}_2 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R}_2^T & \mathbf{0} \\ \mathbf{R}_2^T sk(-\vec{\mathbf{p}}_2) & \mathbf{R}_2^T \end{bmatrix} \cdot \begin{bmatrix} \mathbf{C}_{\vec{\alpha}_1} \mathbf{R}_2 & \mathbf{C}_{\vec{\alpha}_1} sk(\vec{\mathbf{p}}_2) \mathbf{R}_2 \\ \mathbf{0} & \mathbf{C}_{\vec{\epsilon}_1} \mathbf{R}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_2^T \mathbf{C}_{\vec{\alpha}_1} \mathbf{R}_2 & \mathbf{R}_2^T \mathbf{C}_{\vec{\alpha}_1} sk(\vec{\mathbf{p}}_2) \mathbf{R}_2 \\ \mathbf{R}_2^T sk(-\vec{\mathbf{p}}_2) \mathbf{C}_{\vec{\alpha}_1} \mathbf{R}_2 & \mathbf{R}_2^T \mathbf{C}_{\vec{\epsilon}_1} \mathbf{R}_2 \end{bmatrix} \\ \text{cov}(\vec{\eta}_3) &= \begin{bmatrix} \mathbf{R}_2^T \mathbf{C}_{\vec{\alpha}_1} \mathbf{R}_2 + \mathbf{C}_{\vec{\alpha}_2} & \mathbf{R}_2^T \mathbf{C}_{\vec{\alpha}_1} sk(\vec{\mathbf{p}}_2) \mathbf{R}_2 \\ \mathbf{R}_2^T sk(-\vec{\mathbf{p}}_2) \mathbf{C}_{\vec{\alpha}_1} \mathbf{R}_2 & \mathbf{R}_2^T \mathbf{C}_{\vec{\epsilon}_1} \mathbf{R}_2 + \mathbf{C}_{\vec{\epsilon}_2} \end{bmatrix}\end{aligned}$$

JHU Laboratory for Computational Sensing and Robotics



128

Probabilistic Error Modeling: Multivariable Gaussian

Suppose $\bar{\mathbf{x}} \sim N(\bar{\boldsymbol{\mu}}_x, \mathbf{C}_x)$ and $\bar{\mathbf{y}} \sim N(\bar{\boldsymbol{\mu}}_y, \mathbf{C}_y)$

$$\bar{\boldsymbol{\mu}}_{x-y} = E[\bar{\mathbf{x}} - \bar{\mathbf{y}}] = \bar{\boldsymbol{\mu}}_x - \bar{\boldsymbol{\mu}}_y$$

$$\begin{aligned}\mathbf{C}_{x-y} &= E[(\bar{\mathbf{x}} - \bar{\mathbf{y}})(\bar{\mathbf{x}} - \bar{\mathbf{y}})^T] - (\bar{\boldsymbol{\mu}}_x - \bar{\boldsymbol{\mu}}_y)(\bar{\boldsymbol{\mu}}_x - \bar{\boldsymbol{\mu}}_y)^T \\ &= \mathbf{C}_{xx} - 2\mathbf{C}_{xy} + \mathbf{C}_{yy} \\ &= \mathbf{C}_{xx} + \mathbf{C}_{yy} \text{ if } \bar{\mathbf{x}} \text{ and } \bar{\mathbf{y}} \text{ are independent}\end{aligned}$$

Similarly, $\bar{\boldsymbol{\mu}}_{x+y} = E[\bar{\mathbf{x}} + \bar{\mathbf{y}}] = \bar{\boldsymbol{\mu}}_x + \bar{\boldsymbol{\mu}}_y$

$$\begin{aligned}\mathbf{C}_{x+y} &= E[(\bar{\mathbf{x}} + \bar{\mathbf{y}})(\bar{\mathbf{x}} + \bar{\mathbf{y}})^T] - (\bar{\boldsymbol{\mu}}_x + \bar{\boldsymbol{\mu}}_y)(\bar{\boldsymbol{\mu}}_x + \bar{\boldsymbol{\mu}}_y)^T \\ &= \mathbf{C}_{xx} + 2\mathbf{C}_{xy} + \mathbf{C}_{yy} \\ &= \mathbf{C}_{xx} + \mathbf{C}_{yy} \text{ if } \bar{\mathbf{x}} \text{ and } \bar{\mathbf{y}} \text{ are independent}\end{aligned}$$



Probabilistic Error Modeling: Multivariable Gaussian

Suppose $\bar{\mathbf{x}} \sim N(\bar{\boldsymbol{\mu}}_x, \mathbf{C}_{xx})$, $\bar{\mathbf{y}} \sim N(\bar{\boldsymbol{\mu}}_y, \mathbf{C}_{yy})$, and $\bar{\mathbf{z}} = \begin{bmatrix} \bar{\mathbf{x}} \\ \bar{\mathbf{y}} \end{bmatrix}$

$$\text{Then } \mathbf{C}_{zz} = \begin{bmatrix} \mathbf{C}_{xx} & \mathbf{C}_{xy} \\ \mathbf{C}_{yx} & \mathbf{C}_{yy} \end{bmatrix}$$

If we know that $\bar{\mathbf{y}} = \bar{\mathbf{a}}$ then $[\bar{\mathbf{x}}|\bar{\mathbf{y}} = \bar{\mathbf{a}}] \sim N(\bar{\boldsymbol{\mu}}_{[x|y=a]}, \mathbf{C}_{[x|y=a]}),$ where

$$\bar{\boldsymbol{\mu}}_{[x|y=a]} = \bar{\boldsymbol{\mu}}_x + \mathbf{C}_{xy} \mathbf{C}_{yy}^{-1} (\bar{\mathbf{a}} - \bar{\boldsymbol{\mu}}_y)$$

$$\mathbf{C}_{[x|y=a]} = \mathbf{C}_{xx} - \mathbf{C}_{xy} \mathbf{C}_{yy}^{-1} \mathbf{C}_{yx}$$

Also $\bar{\mathbf{y}}$ and $\bar{\mathbf{x}} - \mathbf{C}_{xy} \mathbf{C}_{yy}^{-1} \bar{\mathbf{y}}$ are independent.



Probabilistic Error Modeling: Multivariable Gaussian

Suppose $\bar{\mathbf{x}} \sim N(\bar{\mu}_x, \mathbf{C}_{xx})$, $\bar{\mathbf{y}} \sim N(\bar{\mu}_y, \mathbf{C}_{yy})$ are two different ways to estimate the same quantity, then we can define $\bar{\mathbf{z}} = \bar{\mathbf{x}} - \bar{\mathbf{y}}$

$$\bar{\mu}_z = E[\bar{\mathbf{x}} - \bar{\mathbf{y}}] = \bar{\mu}_x - \bar{\mu}_y$$

$$\mathbf{C}_{zz} = \mathbf{C}_{xx} - 2\mathbf{C}_{xy} + \mathbf{C}_{yy}; \mathbf{C}_{xz} = \mathbf{C}_{zx} = \mathbf{C}_{xx} - \mathbf{C}_{xy}$$

$$\text{Define } \bar{\mathbf{w}} = \begin{bmatrix} \bar{\mathbf{x}} \\ \bar{\mathbf{z}} \end{bmatrix} \quad \text{Cov}(\bar{\mathbf{w}}) = \begin{bmatrix} \mathbf{C}_{xx} & \mathbf{C}_{xx} - \mathbf{C}_{xy} \\ \mathbf{C}_{xx} - \mathbf{C}_{xy} & \mathbf{C}_{xx} - 2\mathbf{C}_{xy} + \mathbf{C}_{yy} \end{bmatrix}$$

Then

$$E[\bar{\mathbf{x}} | \bar{\mathbf{z}} = \bar{\mathbf{x}} - \bar{\mathbf{y}}] = \bar{\mu}_x + \mathbf{C}_{xz} \mathbf{C}_{zz}^{-1} (\bar{\mu}_y - \bar{\mu}_x)$$

$$\text{Cov}[\bar{\mathbf{x}} | \bar{\mathbf{z}} = \bar{\mathbf{x}} - \bar{\mathbf{y}}] = \mathbf{C}_{xx} - \mathbf{C}_{xz} \mathbf{C}_{zz}^{-1} \mathbf{C}_{zx}$$

Note: if $\bar{\mathbf{x}}$ and $\bar{\mathbf{y}}$ are independent, then $\mathbf{C}_{xy} = \mathbf{C}_{yx} = \mathbf{0}$, so

$$E[\bar{\mathbf{x}} | \bar{\mathbf{z}} = \bar{\mathbf{x}} - \bar{\mathbf{y}}] = \bar{\mu}_x + \mathbf{C}_{xx} (\mathbf{C}_{xx} + \mathbf{C}_{yy})^{-1} (\bar{\mu}_y - \bar{\mu}_x)$$

$$\text{Cov}[\bar{\mathbf{x}} | \bar{\mathbf{z}} = \bar{\mathbf{x}} - \bar{\mathbf{y}}] = \mathbf{C}_{xx} - \mathbf{C}_{xx} (\mathbf{C}_{xx} + \mathbf{C}_{yy})^{-1} \mathbf{C}_{xx}$$

